Low-Energy Neutrino Event Reconstruction in DUNE: Clustering, Vertexing, and Energy Reconstruction

Sergio Manthey Corchado

Centro de Investigaciones Energéticas, Medioambientales y Tecnológicas (CIEMAT), Madrid, Spain

October 4, 2025

Abstract

The Deep Underground Neutrino Experiment (DUNE) aims to measure the two most energetic components of the flux of solar neutrinos (^{8}B & *hep*), as well as neutrinos from other astrophysical sources such as nearby supernovae or the diffuse supernovae neutrino background. The sensitivity to the ν–Ar charged current channel, with orders of magnitude higher cross–section than the ν elastic scattering on elements, together with DUNE's high exposure and great energy resolution, present an exceptional advantage in understanding the characteristics of this neutrino flux. This work presents a detailed study of low–energy charged–current neutrino event reconstruction and energy resolution performance in DUNE, under the presence of significant radiological and external backgrounds. Described are the development and validation of clustering algorithms that identify primary ionization signals and associated adjacent energy depositions to account for spatially correlated activity. For the purpose of solar neutrino analyses, this work requires the additional association of charge and light signals. A dedicated energy calibration strategy is employed to recover the interacting neutrino energy. The impact of various background scenarios on the reconstruction is assessed by reporting on the energy resolution achieved across a range of true neutrino energies. Highlighted are the algorithm's effectiveness in maintaining reconstruction fidelity in the low–energy regime. These results demonstrate the robustness of the reconstruction approach and its potential contribution to the solar–ν physics program in DUNE.

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1 Introduction

The Deep Underground Neutrino Experiment, as a next–generation large–scale neutrino detector, will have the ability to shed light on some missing pieces of the solar neutrino puzzle. In particular, DUNE will be sensitive to the two most energetic components of the solar neutrino flux, ^{8}B and *hep* neutrinos, as well as, neutrinos from a nearby occurring supernovae. The high cross–section of Charged Current (CC) interaction on argon nuclei [1], together with DUNE's large active mass and excellent energy resolution, will allow for precision measurements of the astrophysical neutrino flux and their energy spectra. These measurements will provide insights into neutrino properties (such as the MSW effect and potential non-standard interactions), solar physics (including metallicity and core temperature), and potentially new physics beyond the Standard Model.

1.1 DUNE Far Detectors

DUNE will be made of four Far Detector (FD) modules based on the Liquid Argon Time Projection Chamber (LArTPC) technology in the Sanford Underground Research Facility (SURF) to survey a wide portfolio of neutrino physics topics. The FDs will be built in two phases. The first stage will deploy two modules with approximately 14 kt of active mass. Two distinct designs (or geometries), are going to be implemented: the Horizontal–Drift (HD) and the Vertical–Drift (VD) designs (see fig. 1), named after the orientation of their drift fields. The HD's fiducial volume will extend 14.4 m in the drift direction, aligned along the drift coordinate (x), 12 m along the vertical coordinate y, and 58 m along the beam–aligned coordinate (z). This module will be separated into four distinct drift volumes alternating three Anode Plane Assembly (APA) and two Cathode Plane Assembly (CPA) sets, which are set to a drift potential of 180 kV, producing an electric field of 500 V cm^{-1}.

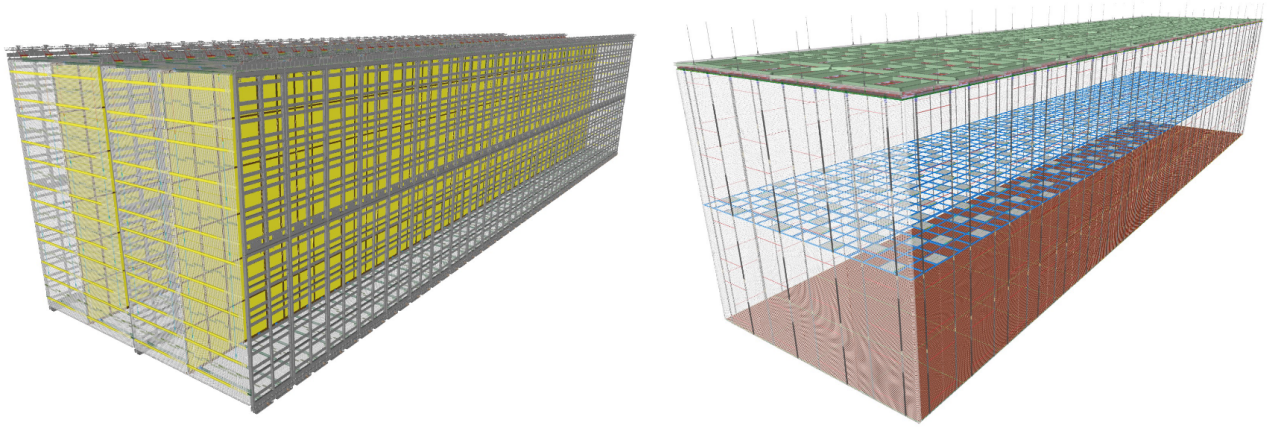


Figure 1: (left) DUNE FD HD module. (right) DUNE FD VD module.

The FD VD module simplifies the traditional HD design by dividing the active mass into just two drift volumes (of dimensions $13.5\times6.5\times60\text{m}^3$) which separated horizontally by the cathode (denoted simply as Cathode Plane). The equivalent of the CPA in the VD volume is called the Charge Readout Plane (CRP) and, differently to the thin and free frame–tensioned wires of a traditional LArTPC, its building block is a perforated Printed Circuit Board (PCB) with flat readout strips. Similarly to the case of the free wires in the HD design, these perforations allow for the drifting electrons to pass by two induction planes, before being collected. Making the comparison of simulation results easier between these two modules, the coordinate system of the VD geometry has been rotated by 90° along the z–axis. This results in the x–axis alignment with the drift direction. Given the nature of the LArTPC technology, in both geometries, the y and z coordinate reconstruction will follow from the Time Projection Chamber (TPC) readout, together with a time–stamp. For beam events, the remaining x coordinate reconstruction follows from a simple calculation of the beam trigger and the already mentioned TPC time–stamp. Note that for non–beam events (such as astrophysical neutrinos), the interaction vertex along the drift volume can only be reliably reconstructed from the TPC charge for APA– or CPA–crossing tracks.

Astrophysical neutrinos accessible to DUNE are three orders of magnitude lower in energy than their beam counterparts and therefore produce tracks that are very small compared to the size of the Liquid Argon (LAr) volume. To improving the reconstruction capabilities of the experiment, both detectors incorporate a Photon Detection System (PDS) which makes use of X–ARAPUCA (XA) modules [2] distributed over the APA (for HD) or cathode and membrane planes (for VD). These provide an additional and instantaneous light signal for any interaction which, provided a correct match from the TPC readout is found, can be used to reconstruct the drift coordinate of low–energy interacting particles. This enables background fiducialization capabilities along the drift direction which would otherwise be impossible for any of the modules.

1.2 Background Model

The measurement of low–energy neutrinos (<20 MeV), particularly those from the Solar ^{8}B chain, requires a thorough understanding of intrinsic (emitted from the LAr or detector components) and external (penetrating from the cavern) backgrounds. One of the most studied backgrounds for LAr detectors is Radon (^{222}Rn), which is a noble gas produced in the decay chain of ^{238}U [3, 4]. As a gas, it can diffuse into the active volume of LAr detectors, where its decay daughters accumulate on surfaces or remain suspended, emitting beta, gamma and alpha radiation (see Appendix 6). Alpha particles have a very short range in LAr, but due to their high energy and quenching, they produce significant light signals in the detector, which can be misidentified as low–energy neutrino interaction candidates in the PDS.

Understanding the contributions from these isotopes is crucial in achieving a correct vertex and energy reconstruction required for low–energy neutrino analyses. Apart from the previously mentioned radon chain, a comprehensive list of background sources has been implemented in DUNE's simulation. These backgrounds can be classified into three main categories: intrinsic, component–induced, and external backgrounds from the cavern.

Presenting DUNE's full background model is out of the scope of this work. Hence, we focus on providing a brief summary while discussing the features that have the biggest impact on the low–energy neutrino reconstruction analyses.

- **External Backgrounds**: External radiation originates outside the detector's cryostat. The flux is divided into two kinds of particles that manage to penetrate into the detector:

  1. **Cavern Neutrons**: Produced by (α, n) reactions in surrounding rock or via cosmic–ray spallation, neutrons are emitted up to ∼10 MeV in kinetic energy. They can enter the active volume where they undergo Elastic Scattering (ES) until they capture producing a gamma–cascade which can mimic the topology of a CC neutrino interaction. Some neutrons can be captured on the cryostat and other detector components contributing to the environmental particle flux with high energy gammas.
  2. **Environmental gammas**: High–energy photons from the rock or shotcrete surrounding the detector, typically originating from natural radioactivity. The highest rate contributor to this flux in DUNE's caver at SURF is the shotcrete covering the walls. Low–energy gammas (<2.7 MeV) entering the active volume have a huge flux contributing mostly to worsen the signal reconstruction by mimicking the de–excitation gammas of the CC interaction or making the topological distinction between ES and CC less reliable. The higher energy external gammas (2.7 < E <14 MeV), are produced by γ–decay of isotopes like ^{208}Tl (∼3.5 MeV) or ^{214}Bi (∼3.27 MeV) and secondary nuclear reactions of (n, γ) interactions in certain geological environments (>6 MeV). These high–energy gamma rays are highly site–specific, depending on the local neutron flux and rock composition. Therefore, in situ measurements are fundamental to assess their impact and model accurately their contribution as fake signals to the different physics analyses.

- **Component-Induced Backgrounds**: These can arise from the materials used to construct the detector itself. Material screening and selection, as well as radon mitigation systems, are essential to reduce these sources. These include decay chains of long–lived isotopes introduced during fabrication or operation. Some of the most prominent are:

  1. **Uranium (^{238}U)**: Produces alpha, beta, and gamma emissions, as well as daughters like ^{214}Bi and ^{210}Pb.
  2. **Thorium (^{232}Th)**: Similar to the uranium chain, with prominent gamma emitters like ^{208}Tl (2.6 MeV).
  3. **Potassium (^{40}K)**: Often present in plastics and glass (like the ones used for traditional photon detectors). It emits a 1.46 MeV gamma line.
  4. **Radon (^{222}Rn)**: Present in atmospheric air and therefore very difficult to prevent.

- **Intrinsic Liquid Argon Backgrounds**: When talking about the properties of DUNE's LAr we are generalizing the properties of (^{40}Ar) but certain radioactive isotopes are either present naturally in atmospheric argon or may be introduced as impurities:

  1. **Argon (^{39}Ar)**: A beta emitter with an endpoint energy of 565 keV and a half–life of 269 years. It dominates the low–energy spectrum in atmospheric argon with an activity of $\sim 1\,\mathrm{Bq\,kg^{-1}}$. Together with the external gammas, this background contributes to the smearing of signals and worsens the CC and ES separation.
  2. **Argon (^{42}Ar) and Potassium (^{42}K)**: Rare isotopes that decay via beta emission. ^{42}K emits an electron ($Q \sim 3.5$MeV), which is particularly problematic due to its similarity to low–energy solar neutrino events.

3. **Krypton (^{85}Kr)**: A beta emitter with a Q-value of 687 keV, introduced via contamination from air or during purification. Often requires cryogenic distillation to remove.
4. **Radon (^{222}Rn)**: In addition to emanation from components, radon can dissolve in liquid argon.

In DUNE, LAr purification and recirculation installations are planned to ensure constant operating conditions. A detailed background model informed by in situ measurements, simulations, and radiopurity assays has been developed and is under constant revision to ensure accurate predictions of the background fluxes in the LAr active volume. The model is implemented in Liquid Argon Software Framework (LArSoft), the software framework used for DUNE's simulations, and is used to generate realistic event samples for detector response studies and physics analyses.

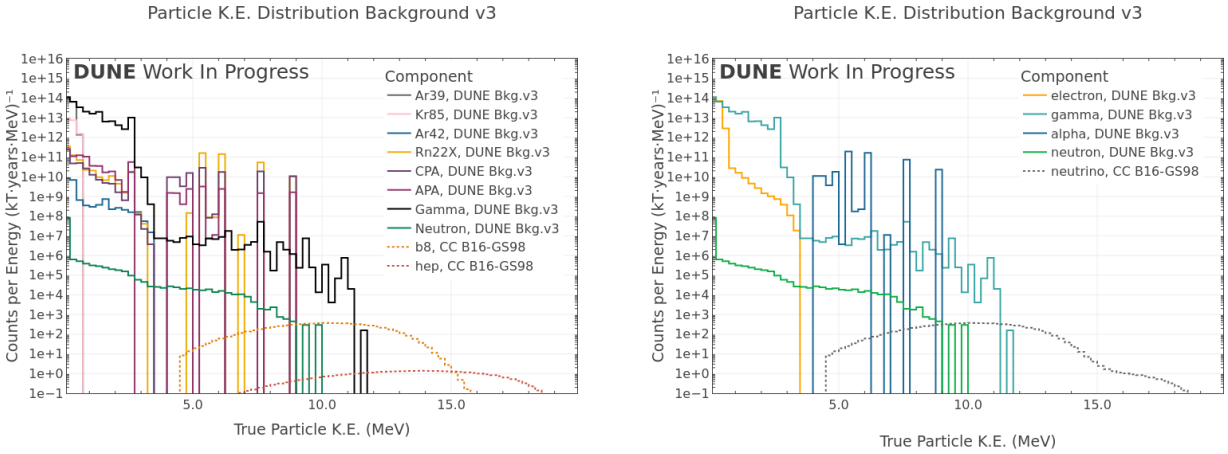


Figure 2: Simulated particle flux from background model v.3 on top of predicted solar spectrum. (left) Divided by the background component. (right) Divided by particle type.

In fig. 2, we show the latest simulated particle flux from the background model (v.3.6) on top of the predicted solar spectrum as a function of the particles' kinetic energy. The left panel shows the flux divided by background components, while the right panel shows it divided by particle type. The flux is dominated by external gammas, argon and krypton at low– and external gammas and neutrons at high–energies. These figures highlight the importance of understanding these backgrounds by highlighting the huge difference in flux between the low–energy neutrino signal and the background components. Without a proper energy calibration, the smearing of the background components can lead to a significant loss of signal efficiency and a degradation of the energy resolution.

2 Event Simulation

In a LArTPC, charged particles ionize the argon atoms and an applied electric field causes the liberated electrons to drift toward a set of readout planes. These readout planes are divided into induction and collection planes and, crucially, they are oriented so that the readout direction of the induction plane is rotated with respect to the one of the collection plane; allowing for a 2D reconstruction by matching time–coincident signals. In DUNE, the collection plane has its readout wires or strips oriented along constant z values, and there are two induction planes providing useful redundancy in the plane matching to prevent the reconstruction of accidental noise coincidence. For all planes, the arrival times and charges of all signal are recorded, allowing for a 2.5D–reconstruction of the interaction $< t, y, z >$. In addition to ionization signals, LAr also produces scintillation light, which can be used for triggering, improved calorimetry, background rejection and timing (reconstructing the drift coordinate x). For low–energy events, efficient charge collection and light detection are crucial for a reliable reconstruction. The neutrino event simulation in DUNE follows the standard low–energy workflows of the LArSoft Monte Carlo (MC) framework [5]. An example of the energy deposition and charge readout of a low–energy CC neutrino interaction in LAr can be seen in fig. 3.

2.1 Geometries

Due to the high computational load of low–energy productions (including backgrounds), the full scale geometries of the FDs are represented in simulation with smaller workspace geometries; designed to provide all possible edge and propagation effects of real neutrino and background interactions but reducing the detector's volume to contain all the signal of a single neutrino event. In this work, we used two geometries: the HD workspace

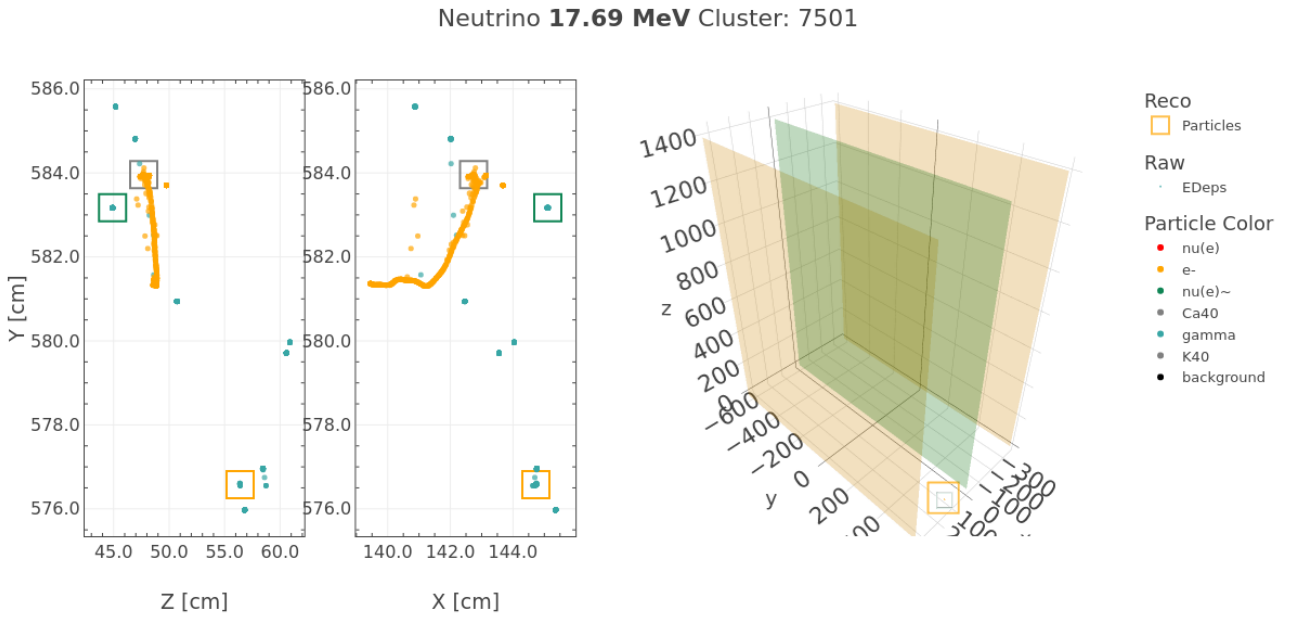


Figure 3: Example of a simulated ∼18 MeV neutrino interaction in the HD geometry. (left) The event displays the two TPC projections of the primary electron energy depositions (orange) and the associated de–excitation gammas (turquoise). The energy deposits from the truth information of the signal propagation are marked with dots and the reconstructed clusters with empty squares. (right) The reconstructed objects are displayed in a 3D representation of the workspace geometry. The represented planes are the central APA (green) and the corresponding CPAs (orange)

geometry (hd_1x2x6) and the VD workspace geometry (vd_1x8x14_3view_30deg). In both cases, the x and y coordinates are set to the same lengths as for the full drift volumes but the z coordinate is shortened roughly 1/4 and 1/3 of the full length, respectively. The HD workspace features a single central APA and two adjacent CPAs creating two drift volumes that are designed to emulate the central region of the HD detector. On the other hand, the VD workspace geometry consists of a single drift volume that represents the upper half of the detector (see fig. 4).

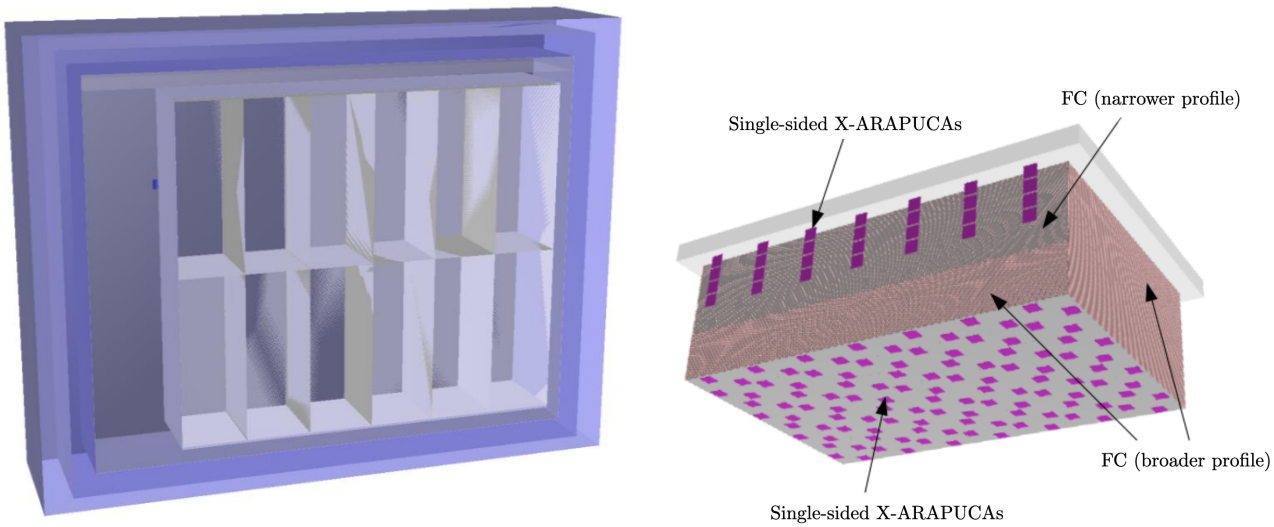


Figure 4: (left) The HD workspace geometry (hd_1x2x6_centralAPA) with a single central APA and a drift length of 3.6 m. (right) The VD workspace geometry (vd_1x8x14_3view_30deg).

The HD geometry is used in low–energy neutrino studies simulating two background scenarios: one without external backgrounds from the lateral faces of the cavern APA (hd_1x2x6_centralAPA) and another one with complete external background (hd_1x2x6_lateralAPA). Due to the arrangement of the TPC planes in the workspace geometry, the latter background scenario is required to implement the lateral cavern flux in the center of the detector (where the APA is placed). This creates a scenario where the central region of the workspace actually

represents the most external face of the detector's charge readout. In both scenarios, we simulate the full set of radiological backgrounds, among with top, bottom, head, and tail external backgrounds.

For the VD geometry, we consider two scenarios: the nominal scenario, which represents the upper half of the VD module (vd_1x8x14_3view_30deg), and a shielded scenario that assumes additional reduction of backgrounds in the form of altered fluxes (vd_1x8x14_3view_30deg_shielded). The latter scenario is likely to represent the lower half of the VD module, where shielding is more suitable from and engineering point of view.

2.2 Neutrino Interaction

The first step in simulating neutrino interactions in DUNE relies on a comprehensive nuclear model of the possible processes. For low–energy events, this is Model of Argon Reaction Low Energy Yields (MARLEY), an event generator developed to simulate the interactions of low–energy neutrinos in the range of a few to tens of MeV. It is particularly suited for simulating interactions relevant to solar and supernova neutrinos in LAr detectors. In this work we focus on the CC process (although ES and Neutral Current (NC) have also been implemented):

$$\nu_e + {}^{40}\mathrm{Ar} \rightarrow e^- + {}^{40}\mathrm{K}^* \tag{1}$$

Upon an electron neutrino capture on an argon nucleus and subsequent electron release, the excitation of the residual ${}^{40}\mathrm{K}^*$ leads to the emission of de–excitation γ–rays, which can produce observable signals (blips) in the vicinity (<1 m) of the main electron track (see fig. 3). MARLEY incorporates a gamma–decay schemes to describe these processes and feeds the resulting daughter particles into LArSoft, which is used to produce a MC input for physics studies. It simulates primary interaction kinematics, the resulting nuclear state and the gamma–cascades from nuclear de–excitation, as well as, other secondary processes (such as neutron or proton emission). In DUNE's physics analyses, the most important signature from these interactions is the already mentioned primary electron track. The resulting energy of these is strongly dependent on the de–excitation scheme of the residual nuclear state (see fig. 5).

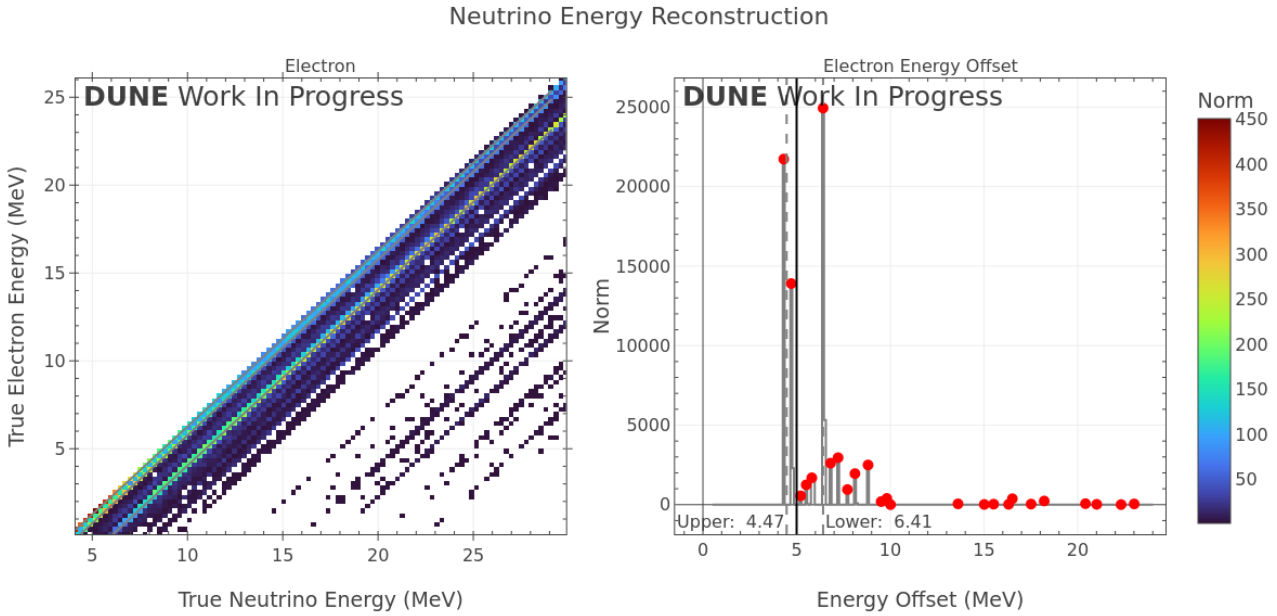


Figure 5: (left) Neutrino–electron energy correlation in a CC interaction as predicted by MARLEY. (right) Gamma energies of the interaction. The solid line splits the energy into two regions with mean values marked with dashed lines.

In fig. 6, the average energy transfer and binding energy of the CC interaction are displayed. One can observe how the gamma channel starts dominating the interaction up to ∼10 MeV, and above that threshold, the main electron takes over the fractional energy. This trend is accompanied by a continuous increase in average kinematic threshold caused by the production of additional secondary particles. The most concerning secondary production is the emission of neutrons, which happens above ∼15 MeV, and leads to a decrease in visible energy that is unlikely to be recovered under realistic reconstruction scenarios. All of these effects combine to a strong energy dependence of the deposited and visible energies, which should be taken into account in our signal calibration. Given that the true neutrino energy reconstruction is our reconstruction goal, the here presented calorimetric algorithms use the number of hits in the primary clusters as a proxy for neutrino energy to encompass the different physics of each energy range.

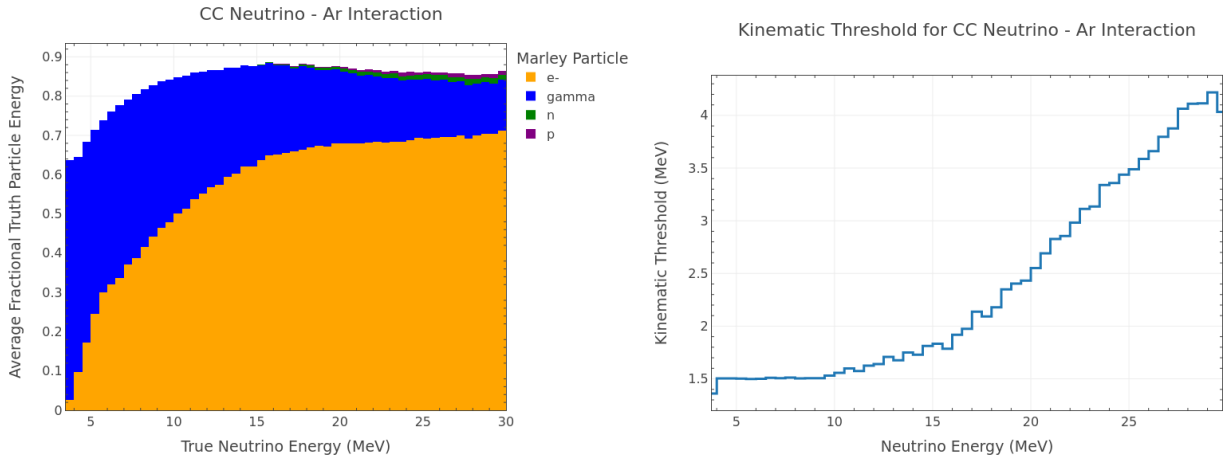


Figure 6: MARLEY's computation of CC neutrino interaction. (left) Daughter fraction of the original kinetic energy. (right) Average kinematic threshold.

It is important to mention at this point that a typical low–energy MC production includes a full propagation of the backgrounds described in section 1.2. These additional particles are produced in their specific rates and concentrations according to computations out of the scope of this publication. All particle kinematics and detector placement happen in the so–called *gen* stage of the simulation. All produced background particles follow the same workflow chain as the signal particles and contribute to the ultimate smearing and resolution of the reconstructed objects. In order to account for all possible matching issues one full drift–time needs to be simulated before the neutrino makes its interaction (at $t = 0$). After this, the simulation last for another full drift–time to allow neutrinos from all regions of the detector to drift and be collected onto the corresponding readout plane. For the purpose of this publication, neutrinos are produced with isotropic positions and directions across the full active volume; and with a flat energy spectrum from 4 MeV up to 30 MeV.

2.3 Particle Propagation

Following the generation of the primary neutrino interaction via the MARLEY generator for low–energy solar neutrino events, the resulting final–state particles are propagated through the liquid argon medium using a Geant4 simulation. This so called *g4* stage, models the detailed particle–matter interactions, including energy loss via ionisation, multiple scattering, bremsstrahlung, photon production, and hadronic interactions; according to a physics list tailored for these low–energy processes.

As charged particles traverse the liquid argon, they lose energy primarily through ionization, producing free ionization electrons and excitation of argon atoms, which subsequently decay and emit prompt scintillation photons in the vacuum ultraviolet (VUV) range (128 nm). These processes are central to the signal generation in a LArTPC. For computational efficiency, the simulation does not explicitly track every ionisation electron or VUV photon. Instead, energy loss along particle trajectories is recorded as a set of energy depositions, or `edeps`, which include position, time, energy, and associated particle identity. Similarly, the number of scintillation photons produced in each energy deposition step is recorded as a secondary output, commonly referred to as `simphotons`. These `edeps` and `simphotons` are then effectively propagated through the LAr volume with custom analytical algorithms in a subsequent stage of simulation.

2.4 Detector Simulation

In this, often called, *detsim* stage; after the effective propagation of ionization electrons and scintillation photons to the TPC (wires or strips) and PDS (photodetectors) readout elements, the signal formation is modelled. This stage takes as input the energy depositions (`edeps`) and the associated scintillation photons (`simphotons`) from the particle propagation step. The detector response to these inputs, produces digitized signals based on the detector's readout electronics and noise characteristics. The two components of the detector response simulation are:

- **TPC Signal**. Ionisation electrons drift under an applied electric field toward the TPC wire planes. During this drift, the electrons experience diffusion (both longitudinal and transverse), attachment to electronegative impurities (parametrised by the electron lifetime), and charge loss due to recombination. These effects are modelled to modify the total collected charge and the temporal and spatial spread of the signal. DUNEs LArTPC use three planes (also called "views") to detect the drifting charge (1 collection plane and 2 induction planes).

1. **Collection plane** responds with unipolar signals as the electrons are finally absorbed.
2. **Induction planes** respond with bipolar signals as electrons pass by without being collected.

The motion of charge induces time–dependent signals (waveforms) on the readsout channels, which are digitized at high frequency. These raw waveforms are then processed using simulated electronic noise based on detector-specific measurements, which is applied to emulate realistic signal-to-noise conditions and electronics shaping. This full set of operations are performed in LArSoft using the Wire–Cell toolkit [6]. To improve charge extraction, the waveforms undergo a **deconvolution procedure**, where the electronics response and noise spectrum are inverted in frequency space to recover a cleaner signal representation. The deconvolved waveforms are integrated over time windows to form localized `hits`, which are stored with time, position, and charge–profile information.

- **PDS Signal**. Simultaneously, the scintillation photons are detected by the PDS, composed of XAs. The `simphotons` from Geant4 are converted into raw signals and integrated in units of Photoelectron (PE) using detector-specific quantum efficiencies and timing models. Similarly to the TPC reconstruction, the PDS response is also emulated with realistic timing resolution, dark noise, and electronic gain amplification. Simulated waveforms are produced for each optical channel, overlaid with noise, and optionally deconvolved or filtered. The resulting waveforms are processed to extract *optical hits* (integrated photon pulses) with associated time and PE information.

2.5 Event Reconstruction

As a final step, after the low–energy simulation steps of generation, propagation and detector simulation of the MC workflow. The reconstruction step for these studies includes a basic clustering scheme that applies both to the TPC and PDS hits generating the final TPC clusters and PDS flashes to be used in the analysis. This *reco* stage typically is subdivided into two sub–steps:

1. *Reco1*: The TPC `hits` are disambiguated[1] and grouped according to tight space and time constrains to form `clusters`, which aim to combine all the charge from localized ionisation signals of single particles. The PDS `ophits` are grouped following the same logic but with a much broader space range to account for the larger spread of scintillation light.

2. *Reco2*: The *Reco1* TPC `cluster` are further grouped into `lowe events` containing one `primary` and any number of found `adjacent clusters`. For each `primary cluster`, a flash–matching algorithm is applied to select a single PDS `flash` object to reconstruct the drift–coordinate of the event.

An example of a simulated neutrino interaction in the HD geometry is shown in fig. 7. The event displays the two TPC projections of the primary electron cluster (orange), the associated de–excitation gammas (turquoise) and uncorrelated background activity (black). The biggest energy deposits from the truth information of the signal propagation are marked with empty squares and the reconstructed clusters with filled squares. Compared to fig. 3, the clusters are now reduced to a point–like data structure, which results from the clustering algorithm.

The clustering algorithms developed for this purpose con be found in the following repositories: (duneopdet & dunereco); as well as the associated producer modules: (SolarOpFlash_module.cc & SolarCluster_module.cc). A custom solar workflow leading to the production of these reconstructed objects has been included in the standard production workflow files: solar_reco_dune10kt_1x2x6.fcl & solar_reco_dunevd10kt_1x8x14_3view_30deg.fcl.

2.6 Event Analysis

For the specific purpose of the solar neutrino studies an analyzer module (SolarNuAna_module.cc) has been written to combine these reconstruction objects with the truth information of the event. This module essentially combines all the information needed to perform the calibration and reconstruction of the signal candidates. The biggest caveat from the analysis point of view is that all clusters and flash information is reduced to a point–like data structure. This is, of course, a choice of design in the analyzer and allows for a simpler handling of the data because all event candidates have the same dimension. In future works, this analyzer should be substituted by the more general LowEAna_module.cc which aims to provide the same event reconstruction while keeping the internal structure of the candidate clusters to allow for better calibration and discrimination strategies.

3 Neutrino Reconstruction Strategy

The calibration and characterization of the CC neutrino signal are based on the true information of the main electron of the interaction. If not otherwise mentioned, the plots in this section are produced from a MARLEY

[1]

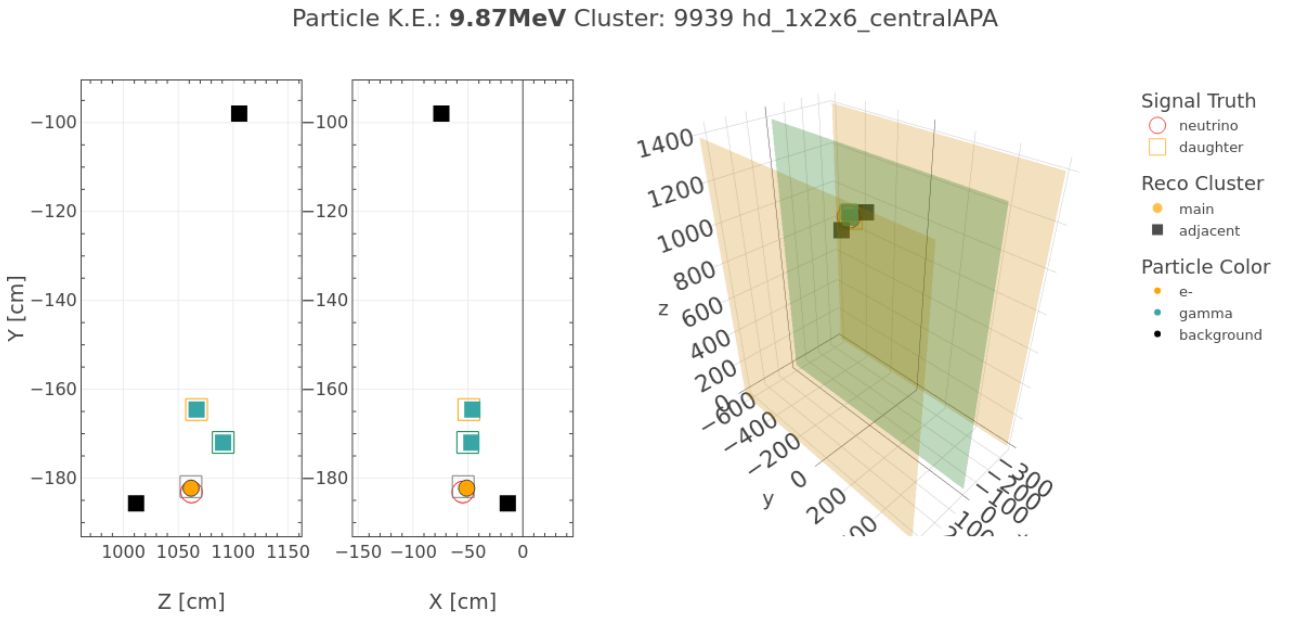


Figure 7: Example of a simulated ∼10 MeV neutrino interaction in the HD geometry. (left) The event displays the two TPC projections of the primary electron cluster (orange), the associated de–excitation gammas (turquoise) and uncorrelated background activity (black). The biggest energy deposits from the truth information of the signal propagation are marked with empty squares and the reconstructed clusters with filled squares. (right) The same points are displayed in a 3D representation of the workspace geometry. The represented planes are the central APA (green) and the corresponding CPAs (orange)

generated CC neutrino sample with central APA background. The first step of the reconstruction process has already happened when clustering the hits of the TPC and PDS detector subsystems in the simulation's *reco1* stage (see section 3.1). Next, we evaluate the detection efficiency and completeness of the collection–induction–matched TPC clusters (see section 3.3). Subsequently, the event electron is corrected in charge for the drift attenuation before computing a hit–dependent calibration factor that recover the particle's original kinetic energy (see section 3.5). Finally, the average calibration factor is applied to all adjacent clusters of the event candidate and the total neutrino energy is computed from four different algorithms (see section 3.7). During the TPC calibration process, the drift coordinate is assumed from truth. A discussion on how the intrinsic background contamination of the detector affects the signal reconstruction at these low energies and the effects on drift coordinate resolution (as a result of TPC–PDS match) are presented in section 4.

3.1 Event Definition

DUNE's solar analysis workflow starts with calibrating the clustered hits from the CC interaction of simulated neutrino events with the MARLEY generator. The CC channel is chosen to be the default based on its dominant cross–section in the low–energy range, as well as its richer topology. All energy and spacial resolution results presented here should be considered and underestimation (or at least compatible) to the ones expected for the ES channel. Each cluster's information is reduced to a point–like object characterized by its number of hits (`nhits`), charge (summed over the charge of the individual hits) and a 2.5D–reconstructed vertex $< t, y, z >$ computed from the charge–weighted average of the individual hits.

Clusters are associated to others based on their relative charges and radial distance. For the radial distance comparison between clusters, the difference in their arrival times is converted to the missing spacial coordinate by taking into account the known drift velocity in LAr. Notice that this computation only holds true for correlated energy deposits (e.g. nuclear de–exitation gammas from a CC neutrino interaction in LAr following the emission of a main electron). In the case of non-correlated background events occurring throughout the full detector volume, their distance along the drift coordinate with respect to other clusters cannot be truthfully determined. From a reconstruction point of view, this leads to the presence of a constant background floor in the vicinity of any signal cluster that corresponds to the projected activity in the interacting drift volume (see table 1 and table 2). These clusters are divided into:

- **primary** with charge bigger than any other's cluster charge up to 1 m radial distance.
- **adjacent** with charge smaller than the one of a primary cluster closer than 1 m radial distance.

Type	Radius (cm)	Mean	STD
Signal	20	0.57	0.77
Background	20	0.06	0.27
Signal	40	0.83	0.96
Background	40	0.22	0.53
Signal	60	0.93	1.04
Background	60	0.43	0.78
Signal	80	0.98	1.07
Background	80	0.66	1.01
Signal	100	1.00	1.09
Background	100	0.84	1.15

Table 1: HD average number of adjacent clusters per primary signal cluster as a function of their 3D–distance for the lateralAPA background scenario.

Type	Radius (cm)	Mean	STD
Signal	20	0.61	0.80
Background	20	0.63	0.84
Signal	40	0.93	1.01
Background	40	2.40	1.80
Signal	60	1.10	1.12
Background	60	5.08	2.94
Signal	80	1.18	1.17
Background	80	8.14	4.12
Signal	100	1.23	1.21
Background	100	10.37	4.93

Table 2: VD average number of adjacent clusters per primary cluster as a function of their 3D–distance for the nominal background scenario.

We define a low–energy event as the combination of a primary cluster together with its adjacent clusters and a list of flashes that are matching candidates for the drift coordinate reconstruction (see 3.2). In any neutrino analysis it will become important to try to distinguish (at least on average) the adjacent cluster distributions for any low–energy event, not only to distinguish background activity (e.g. neutron captures) from signal events, but also to identify the different interaction channels of the neutrino interaction (CC events produce de–exitation gammas that are not present in ES interactions). Consequently, the higher the projected background activity in any of the isolated drift volumes of DUNE's FDs, the harder particle identification and channel separation will become. Notice that there is nothing preventing a single neutrino interaction from producing more than one low–energy event. Rarely secondary gammas can scatter further than 1 m in LAr or a background–induced miss–match in the TPC reconstruction could lead to blips appearing in uncorrelated regions of the detector.

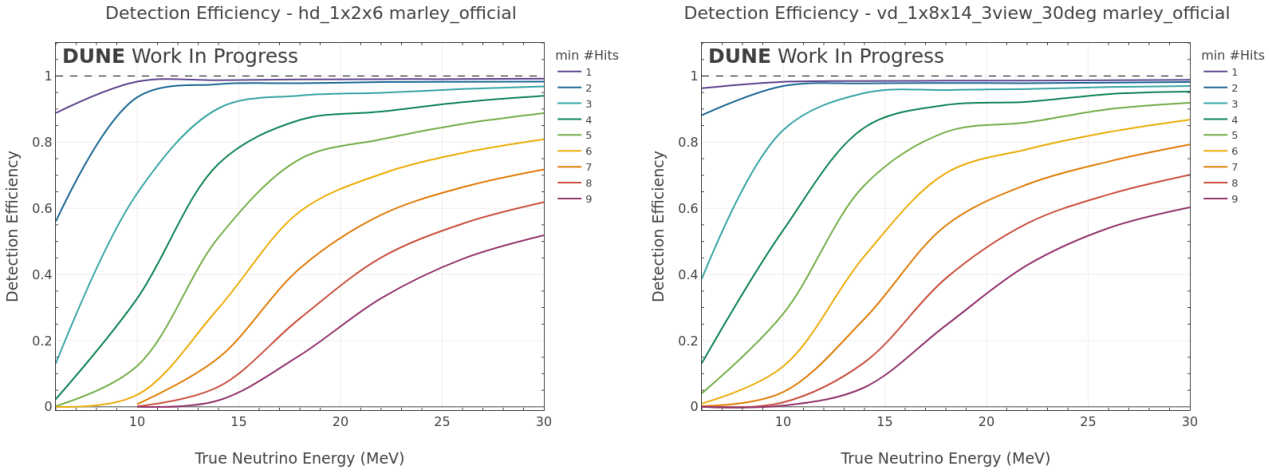


Figure 8: HD (left) and VD (right) detection efficiency for signal clustering parameters of 3 adjacent channels and 12.5 µs.

For the purpose of this work, secondary low–energy events coming from a single neutrino interaction will always be ignored. This is to ensure that the reconstruction performance is evaluated for the most complete reconstruction object in any interaction but in any physics analysis these should be kept and treated as part of the signal or even background.

3.2 Flash Matching

The first step in the signal reconstruction process is the attempt to match the reconstructed primary TPC cluster with its corresponding PDS flash. Note that this step is not strictly required for the energy reconstruction, as the drift attenuation correction can be applied statistically. However, the matching is essential for the correct reconstruction of the neutrino position and, most importantly, background fiducialization. This is particularly important for low–energy solar neutrino analyses, where the signal rate is low and the high energy tails of the background spectra can leak into the signal region. The matching is also important for supernova neutrino analyses, where the time structure of the neutrino burst provides valuable information about the supernova

dynamics.

The matching is performed by evaluating the distance between the TPC vertex and the PDS flash position, as well as the time difference between the TPC cluster and the PDS flash. The matching efficiency is defined as the fraction of TPC clusters that have a corresponding pure PDS flash. fig. 9 shows the flash–matching efficiency as a function of the distance to the PDS plane and the neutrino energy. The matching efficiency is high for distances up to 1 m and decreases for larger distances. The matching efficiency is also higher for higher neutrino energies. This is expected, as the PDS flashes are more likely to be detected for higher energy depositions.

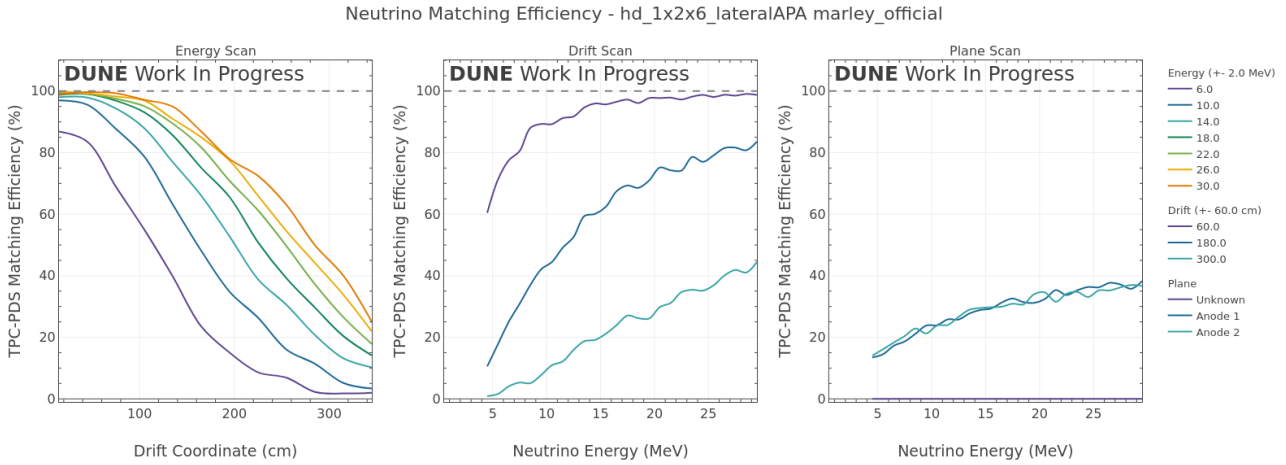


Figure 9: Flash–matching efficiency. (left) As a function of distance to the PDS plane. (right) As a function of neutrino energy.

For any given TPC cluster, the matching strategy follows these steps:

1. **Distance Calculation**: Compute the distance between the TPC cluster vertex and all PDS flash positions.
2. **Time Difference Calculation**: Calculate the time difference between the TPC cluster and the PDS flash. This must be smaller than the drift time of the electrons in the LAr.
3. **Candidate Selection Criteria**: A PDS flash is considered adjacent to a TPC flash if the distance $(\mathrm{TPC_y} - \mathrm{PDS_y})^2/\mathrm{AdjOpFlashY}^2 + (\mathrm{TPC_Z} - \mathrm{PDS_Z})^2/\mathrm{AdjOpFlashZ}$ is smaller than 1, where AdjOpFlashY and AdjOpFlashZ are the adjusted matching distances in the 2 projected coordinates, respectively.
4. **Matching Criteria**: The algorithm consider 3 additional cuts to prevent the random matching of background cluster:
   - The minimum number of hits in the PDS flash: AdjOpFlashMinNHitCut (by default 3).
   - The minimum number of PEs in the PDS flash: AdjOpFlashMinPECut (by default 20).
   - The maximum ratio of AdjOpFlashMaxPE / AdjOpFlashPE: AdjOpFlashMaxPERatioCut (by default 1). This is intended to prevent the matching of background PDS flashes, wich are less homogeneous in the number of PEs compared to the signal PDS flashes.

Select the PDS flash with the highest number of PE as the matching candidate for the TPC cluster. If no PDS flash is found within the matching criteria, the TPC cluster is considered unmatched.

From this matching one can reconstruct the drift coordinate of the TPC cluster, which is essential for the correct reconstruction of the neutrino energy. The drift coordinate is computed from the time between the TPC vertex and the PDS flash by taking into account the drift velocity of the electrons in the LAr.

3.3 Detection Efficiency

The first figure of merit to be extracted from the simulation is the detection efficiency (see fig. 8 (left)). We observe a strong dependence on the number of hits. Only with a minimum hit condition of 2, the simulation reaches detection efficiency values of ∼100%. Achieving a reconstruction threshold (>90% efficiency) above 5.5 MeV requires a minimum hit condition of 1 hit. Instead of interpreting this efficiency in terms of the neutrino electron, using the information provided in fig. 5 (neutrino–electron energy offset >4.5 MeV) one can deduce an electron detection threshold of ∼1 MeV. A better estimation of the detection threshold can be extracted from the energy resolution fit in fig. 25.

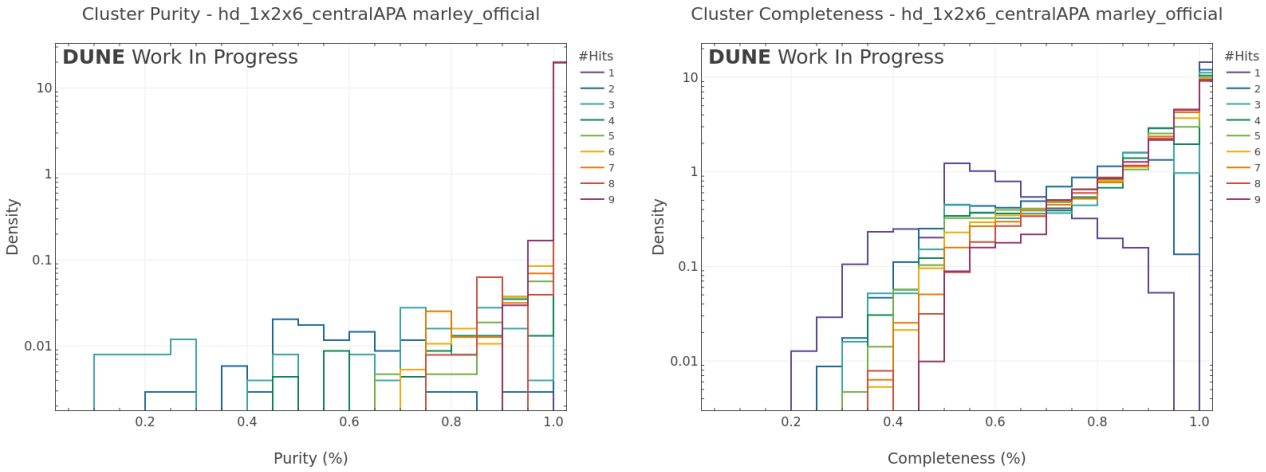


Figure 10: Clustering efficiency plots. (left) Neutrino signal purity in the main cluster. (right) Electron charge completeness in the main cluster.

The next parameter to evaluate is the completeness of the main clusters, which is a measure of the electron charge–confinement given a certain clustering parameter set. For the purpose of the solar neutrino analysis of DUNE, clustering parameters of 3 adjacent channels and 25 ticks have been used to maximize completeness while ensuring cluster purity in the context of background. The most up–to–date values for this clustering should be found in the producer's SolarCluster.fcl. Despite the good overall completeness of the signal, we observe a trend with respect to the number of hits per cluster, which can be accounted for in the later described charged correction factor to improve reconstruction. The trend that one observes is clear and understandable; for depositions which result in a lower number of hits (this can happen because of low energy deposition or a perpendicular trajectory of the particle with respect to the TPC plane) it is more likely that clusters of similar sizes are formed (e.g. 2 clusters of 1 hit) resulting in a lower charge in the main cluster.

3.4 Electron–Lifetime Correction

TPC hits are first grouped into spatial clusters using a dedicated clustering algorithm. Each event contains a **main cluster**, corresponding to the primary outgoing electron, and zero or more **adjacent clusters**, interpreted as gamma-induced "blips". The first step in calibration involves correcting the total ionization charge of each cluster for the electron–lifetime attenuation in the LAr (see section 3.4).

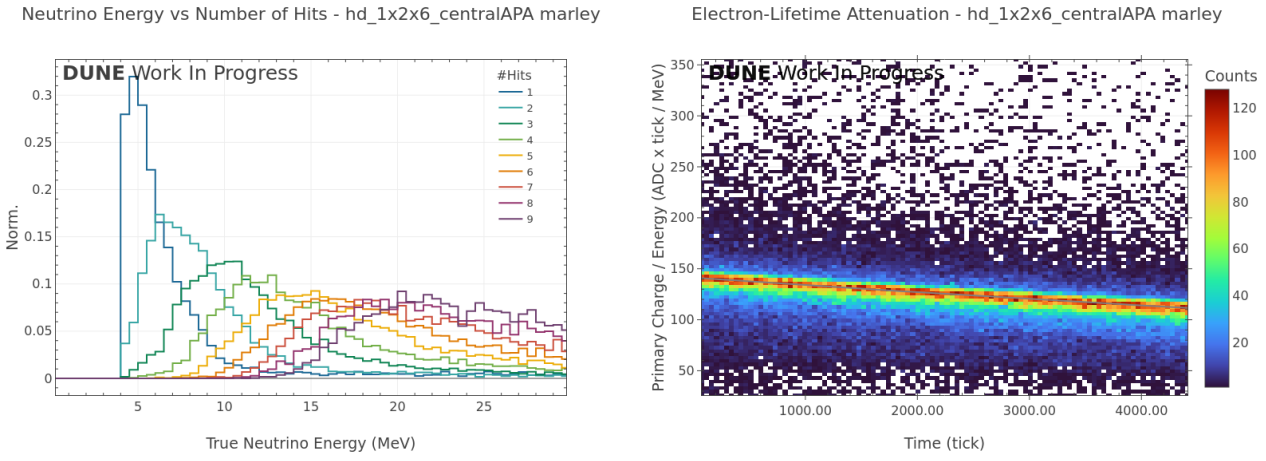


Figure 11: (left) Neutrino energy correlation with number of hits in the primary cluster. (right) Drift attenuation for the main electron cluster.

$$\frac{Q_{\text{raw}}(n_{\text{hit}})}{E_e^{\text{truth}}} = F_0(n_{\text{hit}}) \cdot \exp\left(\frac{t^{\text{drift}}}{\tau_e}\right), \tag{2}$$

where Q_{raw} is the observed (uncorrected) cluster charge, E_e^{truth} is the true main electron energy of the interaction, F_0 is the charge–to–energy correction factor, t^{drift} is the true drift time of the electron, and τ_e is the electron lifetime measured here independently and cross-checked against the simulation input.

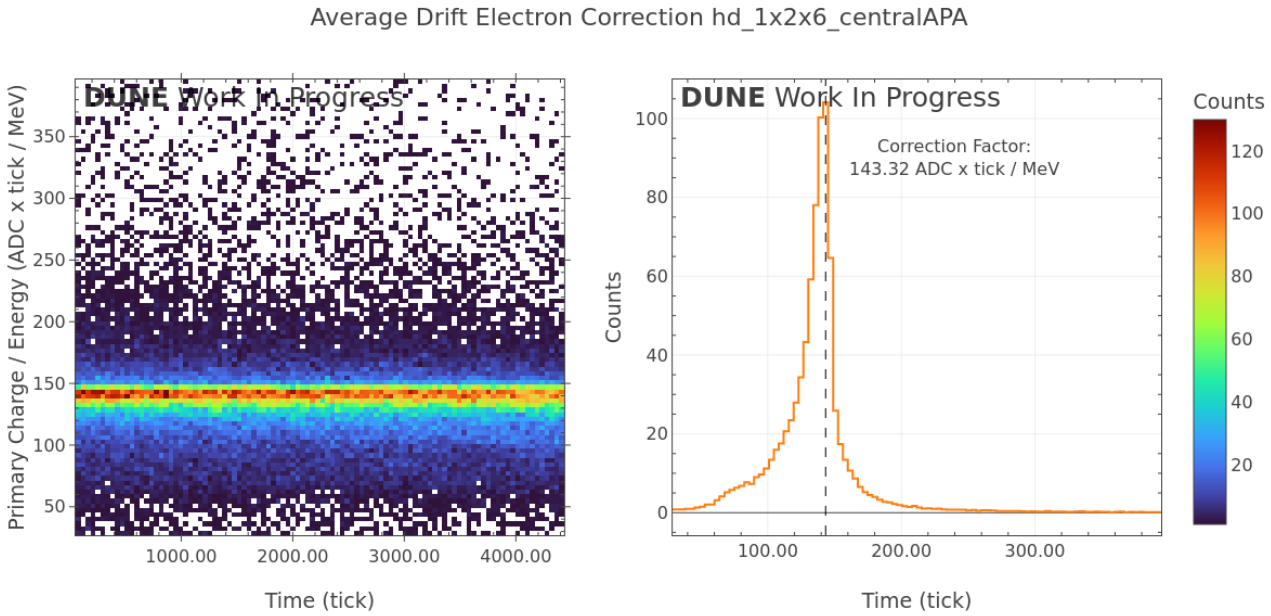


Figure 12: Drift attenuation correction.

Assuming the main cluster corresponds to a fully-contained electron, we take its known true energy (from simulation) as a reference and compute the charge–to–energy conversion factor for each (n_{hit}) threshold. Additionally, to assess the impact of this assumption, with the goal of characterizing signal clusters in the context of background which limits the clustering parameters, one can also investigate a "cheated charge". This is computed for all clusters, adding the surrounding charge that has been backtracked to the main electron of the interaction. A calibration based on this cheated clustering gives us an understanding of the losses that arise from a limited clustering ability in the context of background.

$$E_{\text{cluster}} = \alpha(n_{\text{hit}}, E_{\text{true}}) \cdot Q_{\text{corr}}. \tag{3}$$

After the truth drift correction has been applied to the data, the charge correction factor is immediately obtained.

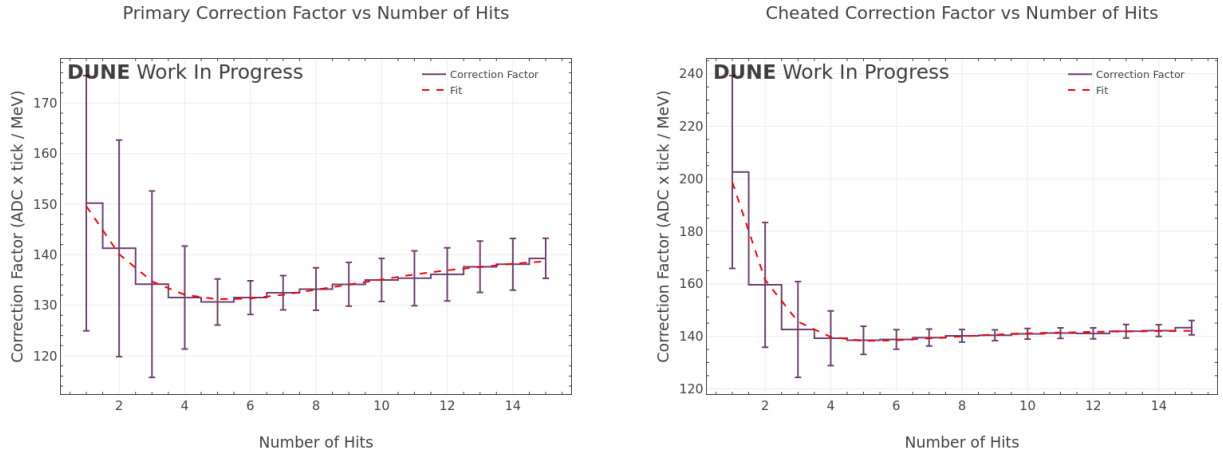


Figure 13: Per hit correction factor distributions. (left) Primary cluster charge. (right) Cheated electron cluster charge.

3.5 Electron Energy Calibration

Even after lifetime and charge calibration, small systematic biases may remain in the reconstructed electron energy due to recombination modelling, imperfect response, or clustering inefficiencies. We apply a linear correction to account for this residual offset:

$$E_e^{\text{final}} = (E_{\text{cluster}} - n)/m, \tag{4}$$

where m and n are derived from the linear fits to the simulated E_{true} versus E_{reco} distributions as shown in fig. 14. The calibration is performed separately for the main electron cluster and the cheated electron cluster. The calibration factor is computed as a function of the number of hits in the cluster (see fig. 15), which serves as a proxy for the electron energy.

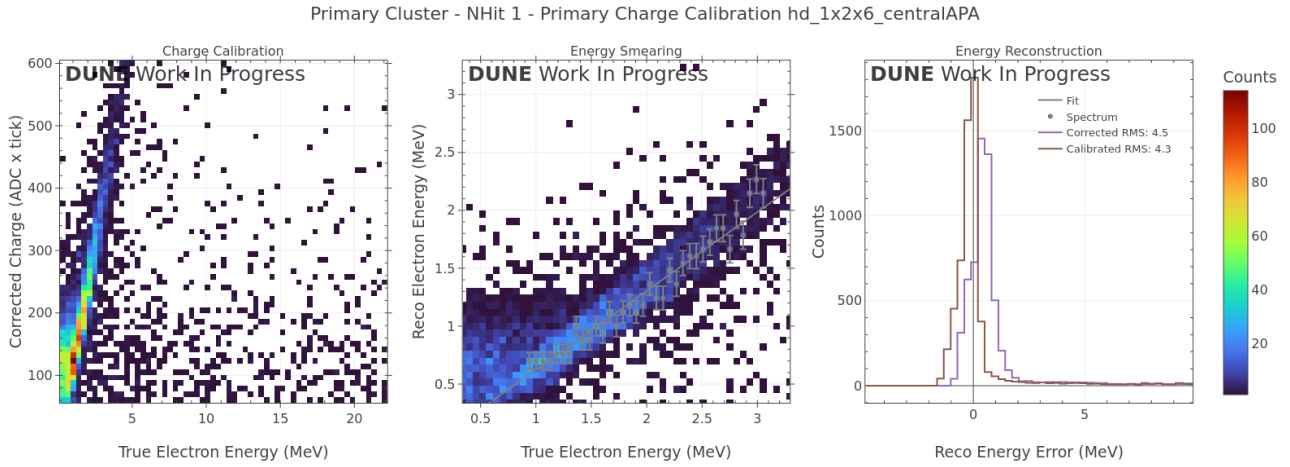


Figure 14: Charge calibration for the main electron cluster. (left) Corrected charge over true electron energy. (center) Linear fit to reco energy over true electron energy. (rigth) Comparison between (true - reco) for corrected (before linear fit calibration) and calibrated (after linear fit calibration) reco energy distributions.

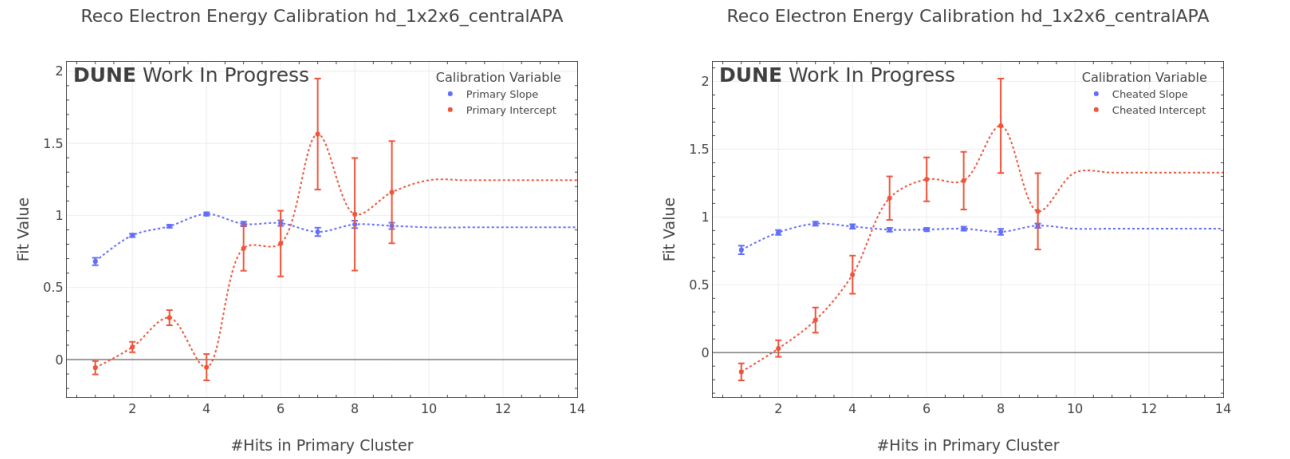


Figure 15: Per hit calibration results. (left) Primary cluster charge. (right) Cheated electron cluster charge. Blue shows the slope and red the intercept values of the linear fits.

3.6 Gamma Energy Calibration

3.7 Neutrino Energy Calibration

The main electron energy is corrected for the drift time attenuation and the charge calibration factor, as described in section 3.5. The main electron energy is then used to compute the total neutrino energy, which is computed from the main electron energy and the energies of all adjacent clusters (up to 1 m). The adjacent clusters are assumed to be gamma-induced "blips" and are calibrated using the same charge calibration factor as the main electron. The total neutrino energy is computed as:

$$E_{\nu}^{\text{raw}} = E_{e}^{\text{final}} + \sum_{i} E_{\gamma,i}^{\text{final}}. \tag{5}$$

This strategy is the first and most obvious approach for neutrino energy reconstruction (**TotalEnergy**). However, it does not account for the energy smearing of the signal, and the presence of background can significantly affect the reconstructed neutrino energy. Therefore, we study alternative methods to improve the energy resolution and mitigate the effects of background contamination.

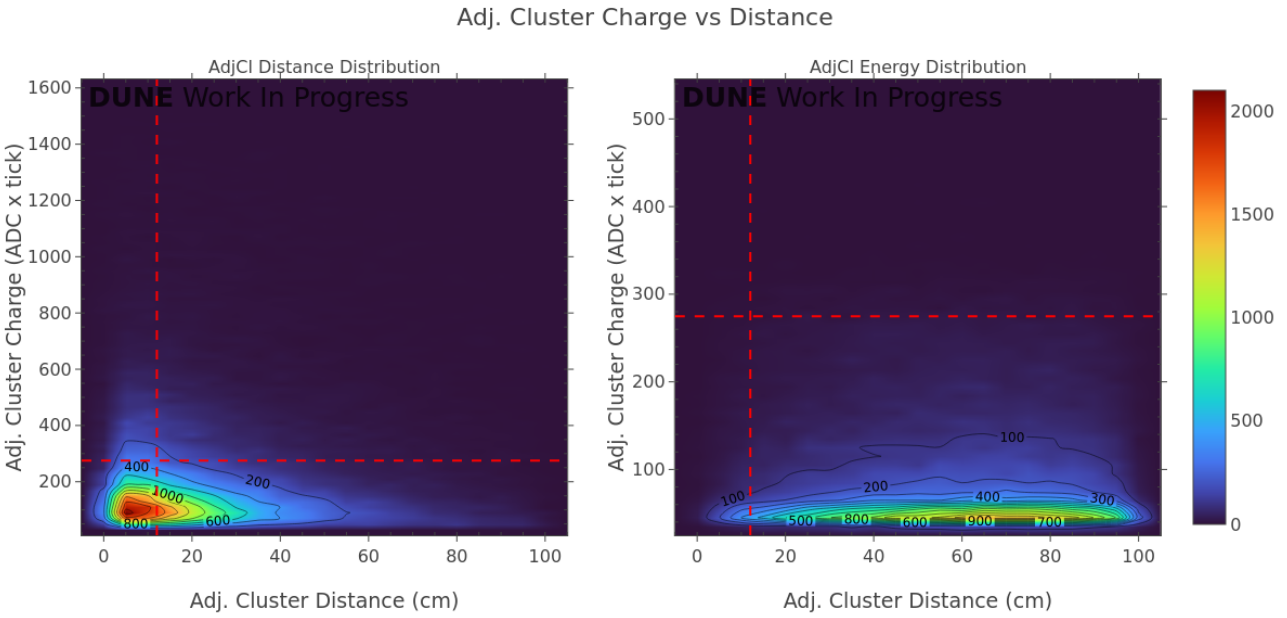


Figure 16: Distribution of the adjacent clusters charge and distance to the main electron cluster.

The next simple solution is to apply a selection on the adjacent clusters (**SelectedEnergy**), which are assumed to be gamma-induced "blips". The selection is based on the adjacent cluster charge and distance to the main electron cluster. The adjacent clusters are selected if they have a charge above a certain threshold and are within a certain distance to the main electron cluster. The thresholds are tuned to maximize the signal over background ratio. fig. 16 shows the distribution of the adjacent clusters charge and distance to the main electron cluster and selection thresholds.

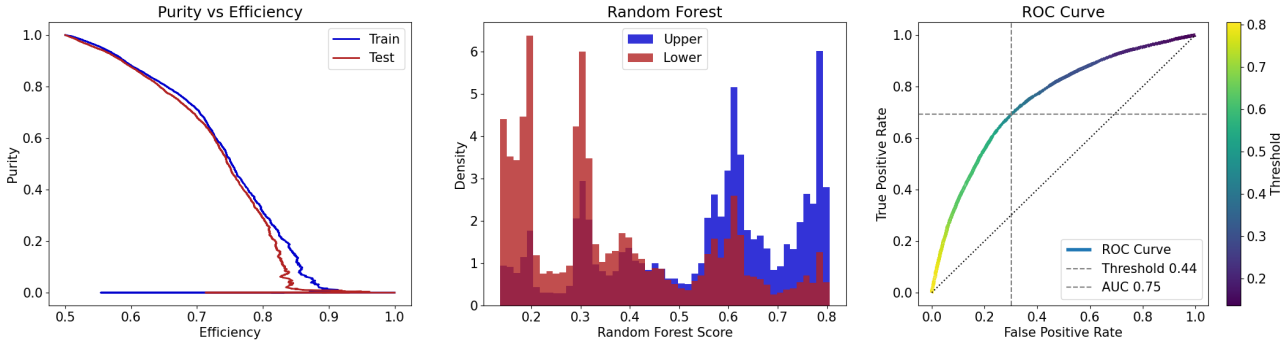


Figure 17: (left) Purity vs Efficiency curve of the random forest regressor for both the train and the test samples. (center) Score of the random forest regressor for both CC samples. (right) ROC curve of the random forest regressor.

The next step is to apply a machine learning algorithm to the adjacent clusters to improve the energy resolution. The algorithm is trained on the main cluster properties and the average adjacent cluster features. The algorithm is a random forest regressor, which is trained on the simulated data. The random forest regressor is used to predict the energy of the adjacent clusters, which are then added to the main electron energy to obtain the total neutrino energy (**SolarEnergy**). To simply the problem and motivated by the two main calibration lines from the CC interaction displayed in fig. 6, the Machine Learning (ML) algorithm is trained two add one of two total adjacent cluster energy values: 4.47 MeV and 6.41 MeV. fig. 17 shows the score of the random

forest regressor which provides an AUC of 0.77 for the test sample, which means that the algorithm is able to distinguish between signal and background with a good efficiency. The random forest regressor is trained on the following features:

- NHits: number of hits in the main electron cluster.
- AdjClNum: number of total adjacent clusters.
- SelectedAdjClNum: number of selected adjacent clusters.
- Energy: energy of the main clusters.
- **TotalAdjClEnergy**: energy of the total adjacent clusters.
- SelectedAdjClEnergy: energy of the selected adjacent clusters.
- SelectedAdjClEnergyRatio: energy ratio of the selected to the total adjacent clusters.
- **MaxAdjClEnergy**: maximum energy of the total adjacent clusters.
- SelectedMaxAdjClEnergy: maximum energy of the selected adjacent clusters.
- TotalEnergy: reconstructed **TotalEnergy**.
- SelectedEnergy: reconstructed **SelectedEnergy**.

From which the highlighted features are the most important for the random forest regressor, as shown in fig. 18.

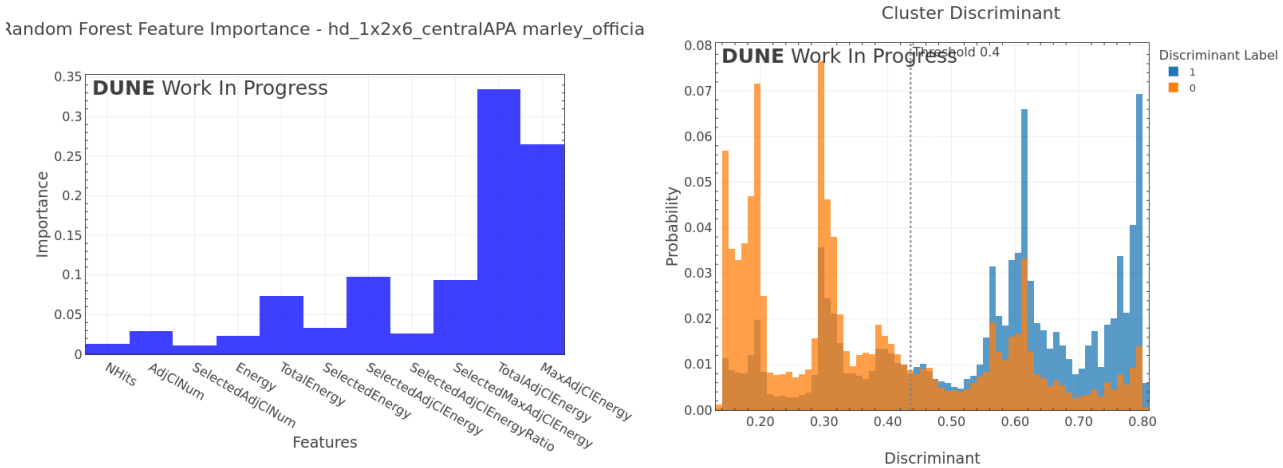


Figure 18: (left) Feature importance of the random forest regressor. (right) Discriminant calibration for the random forest regressor.

Finally, as a benchmark for comparison between the here presented reconstruction algorithms and with the aim of quantizing the impact of adjacent gammas on the total neutrino energy reconstruction, we introduce the **ClusterEnergy** algorithm as the reconstruction method based purely on the main electron cluster. The only correction is a fixed calibration offset applied to the full sample to align the center of the distribution with the true energy.

3.8 Neutrino Energy Smearing

Due to the missing nuclear binding energy and undetected soft particles, the reconstructed neutrino energy is expected to systematically underestimate the true value. As a final step to the reconstruction process, all energies (**TotalEnergy**, **SelectedEnergy**, **SolarEnergy** and **ClusterEnergy**) are calibrated with a linear fit to compensate for systematic losses in the previous clustering or calibration processes. This final global correction is applied to bring the distribution into agreement with the simulation truth:

$$E_\nu^{\text{cal}} = \left(E_\nu^{\text{raw}} - n\right)/m, \tag{6}$$

with m and n obtained from a linear fit to the true versus reconstructed neutrino energy spectrum of the MC sample as shown in fig. 19.

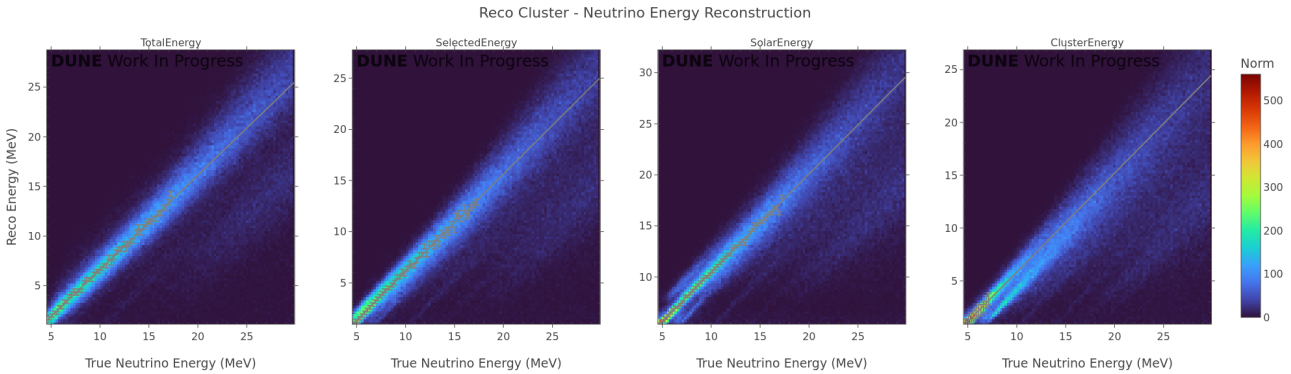


Figure 19: Neutrino energy smearing from the calibrated signal.

4 Results

From the TPC cluster vertex reconstruction and the flash–matching, we can extract the energy and vertex resolution. The calibrated neutrino energy is obtained from a standalone code SOLAR package, which formats the simulation output into numpy arrays and performs the successive operations presented in section 3. Here we present the main figures of merit and compare the resolution performance between the two geometries, the hd_1x2x6_centralAPA and the vd_1x8x14_3view_30deg and additional background reduced scenarios.

4.1 Neutrino Vertex Resolution

The vertex resolution is defined from the distance between the reconstructed vertex and the true vertex position. Note that this analysis reconstructs particle interactions as point–like objects and therefore reconstructs a vertex that is shifted on average to the center of the true particle track. For the cases in which a coordinate cannot be determined (applicable to x– and y– coordinates), the reconstructed coordinate is set to the center of the active volume. The vertex error is then defined as the difference between the reconstructed and true coordinates.

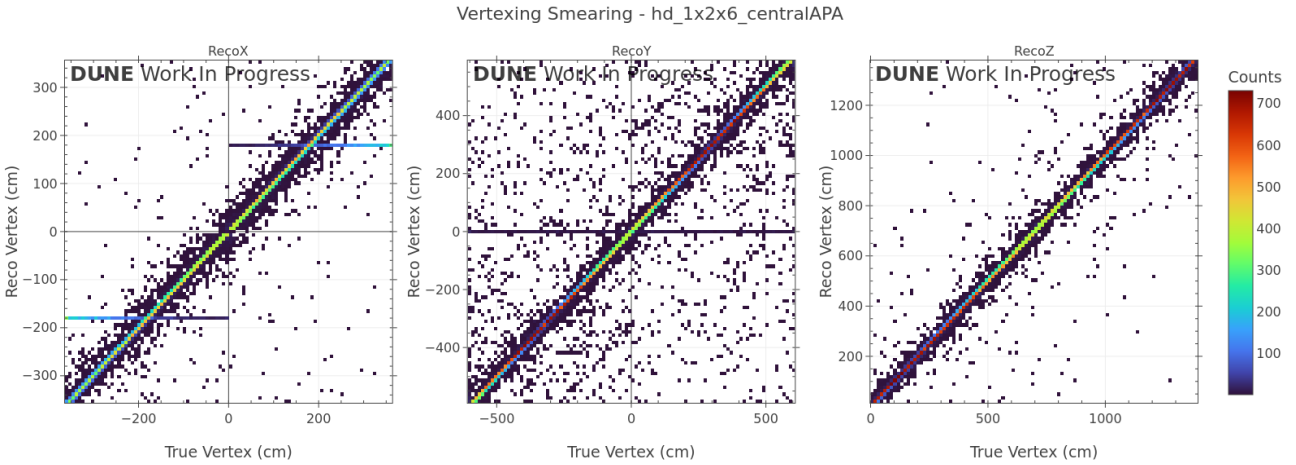


Figure 20: Vertex smearing from calibrated signal.

As explained in section 3, the drift–coordinate of the reconstructed vertex is obtained from the flash–matching algorithm, which combines the information from the TPC cluster and the associated PDS flash. The results are shown in fig. 20 for the *vd_1x8x14_3view_30deg* geometry with nominal background production. These distributions are not Gaussian, as expected from the nature of the matching algorithms. The resulting matching can be correct, leading to a small error, or completely wrong, leading to a large error in the reconstructed coordinate. To have a better understanding of the reconstruction performance, we can compute the cumulative vertex error fig. 21. This plot shows the percentage of events that are reconstructed within a certain vertex error tolerance. For instance, we can see that approximately 90% of the events have a vertex error below 1 cm in the y– and z–coordinates, while only about 60% of the events achieve this level of accuracy in the x–coordinate.

Given that the distributions are not Gaussian, we can still attempt to fit a Gaussian function to the vertex error distributions by ignoring the tails to extract a vertex resolution from the FWHM of the fit as a figure of

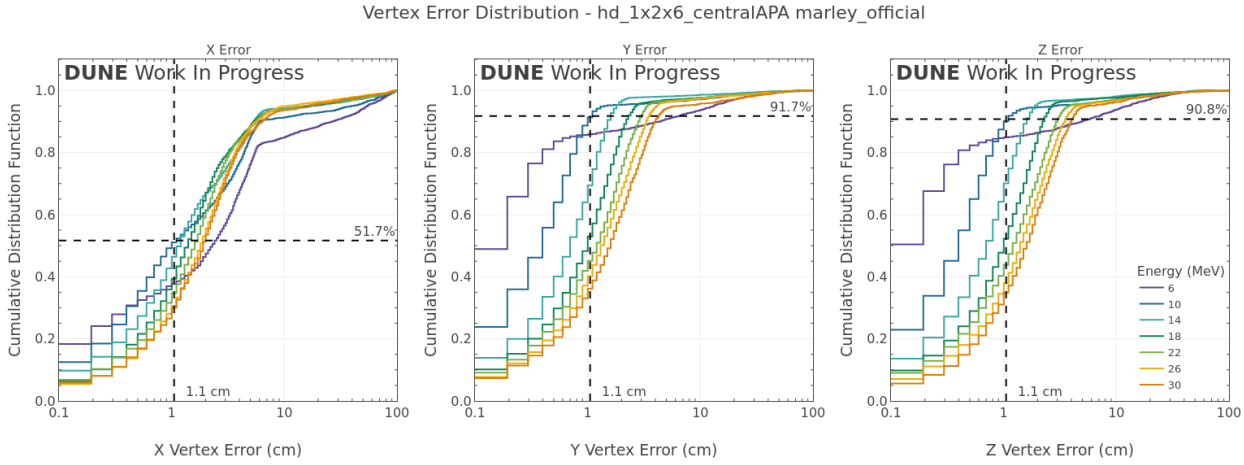


Figure 21: VD cumulative vertex error distribution.

merit. This is done by fitting a Gaussian function to the core of the distribution. The results are shown in fig. 23 for the *vd_1x8x14_3view_30deg* geometry with and without nominal background production.

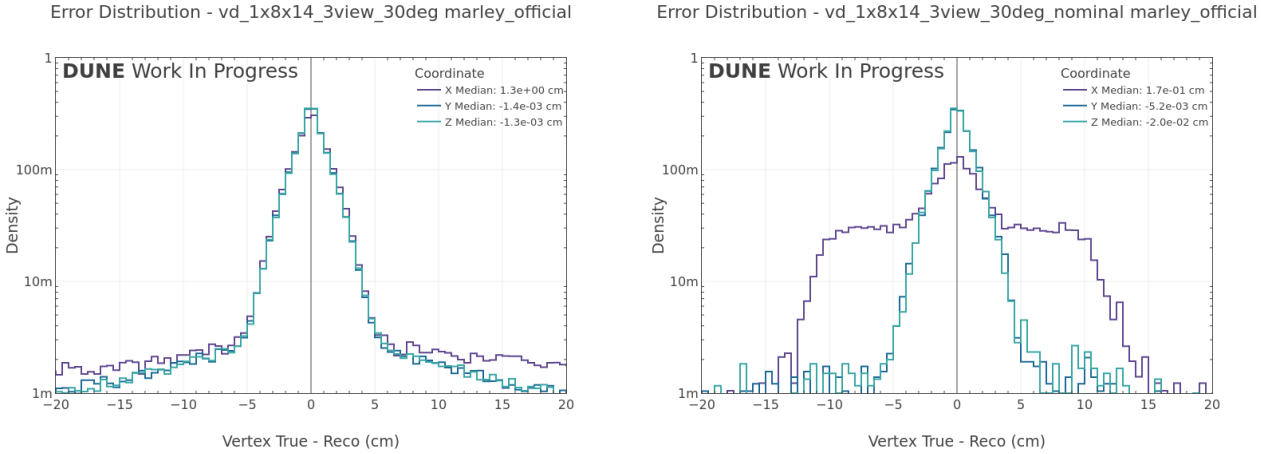


Figure 22: VD vertex error distribution. (left) Neutrino production. (right) Neutrino + nominal background production.

The results of this fitting procedure as a function of true neutrino energy are presented in fig. 23 for both the neutrino-only and neutrino-plus-background samples. The vertex resolution is found to be between 0.5 cm and 2.5 cm except for the x–coordinate in the neutrino-plus-background sample, which is between 1.5 cm and 3.5 cm. This degradation in the x–coordinate resolution is expected due to the pile–up of background events, which leads to a smearing of the time resolution of individual ophits in the correctly associated flashes. The degradation in vertex resolution with increased neutrino energy is a side effect of the point–like reconstruction approach, which places the reconstructed vertex at the center of the neutrino-induced particle track. As the neutrino energy increases, the average particle track size also increases, leading to a larger displacement between the true interaction vertex and the reconstructed vertex. Currently, there is no low–energy physics analysis in DUNE that would strongly benefit from a vertex resolution better than a few centimeters. For example, in the case of solar neutrinos, where external cavern–gammas are one of the major backgrounds, the Compton scattering length is on the order of tens of centimeters, which is much larger than the achieved vertex resolution. But, if needed, the vertex resolution could be improved by implementing a more sophisticated reconstruction algorithm that takes into account the topology of the event.

4.2 Fiducialization Efficiency

In order to reduce the impact of background events originating from the detector materials and the surrounding environment, a fiducial volume cut is typically applied to the reconstructed neutrino events in low energy–analyses. This cut is based on the reconstructed position of the primary TPC cluster associated with the neutrino interaction. The fiducial volume is defined as a region within the detector that is at least x cm away from any

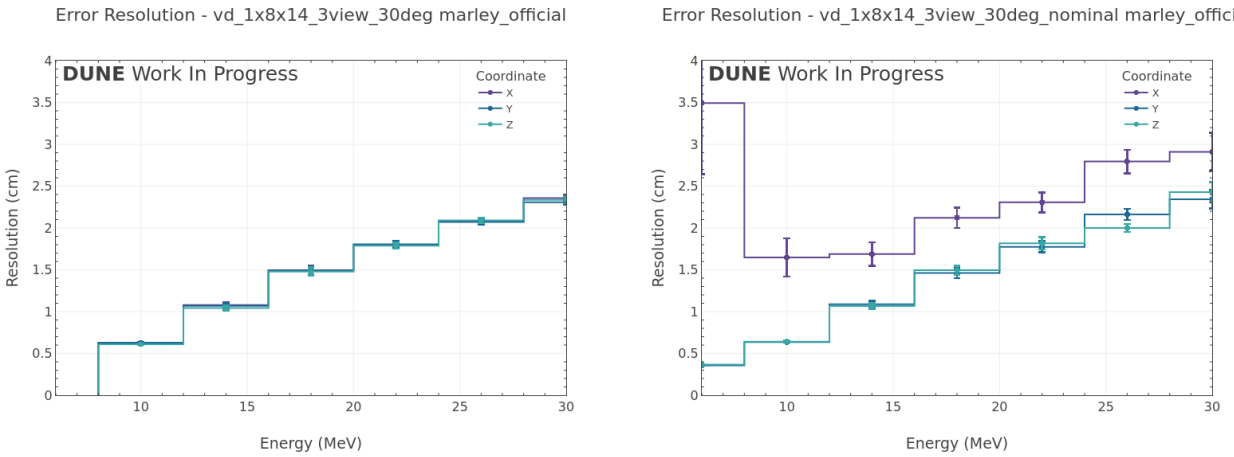


Figure 23: VD vertex resolution from Gaussian fit of error distributions. (left) Neutrino production. (right) Neutrino + nominal background production.

external detector boundary. This distance can be chosen in any analysis to optimize the signal-to-background ratio.

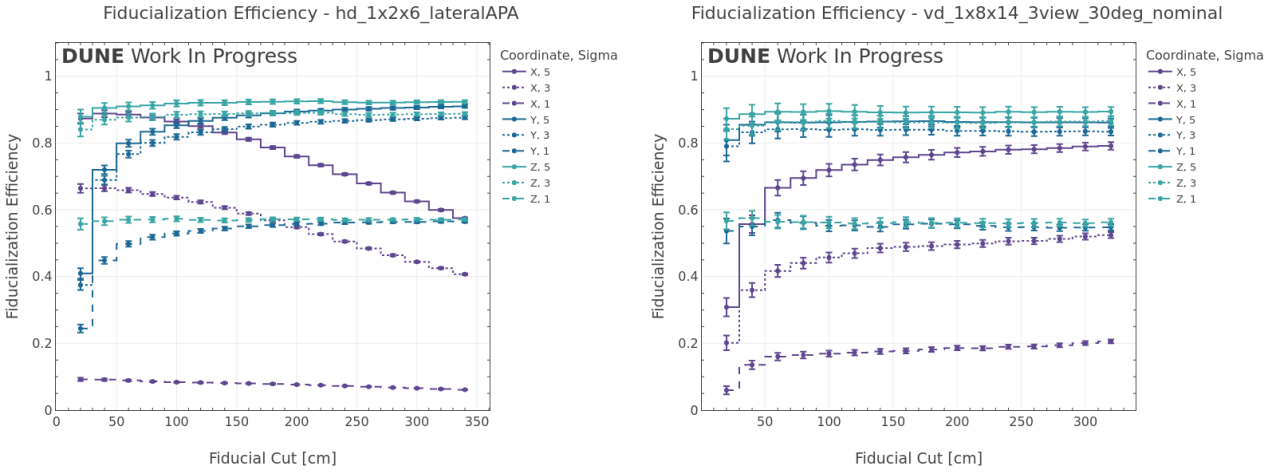


Figure 24: Fiducialization efficiency as a function of the distance to the detector boundary for (left) HD and (right) VD.

4.3 Electron Energy Resolution

As a figure of merit for the process, we can compute the energy resolution from the cheated calibrated energy of the main electron. This is typically the value that analyzers compute when estimating the energy resolution of a TPC. It is obtained for an idealized clustering with ad hoc calorimetry and a high hit threshold (in this case we use 3 for a typical track). The energy resolution is computed as the Root Mean Square (RMS) of the relative error of the reconstructed variable versus the truth. The results are shown in fig. 25. The formula for the energy resolution fit is:

$$\Omega_{(E_{\text{true}})} = \sqrt{\left(\frac{p_2}{\sqrt{E_{\text{true}} - b}}\right)^2 + \left(\frac{p_1}{E_{\text{true}} - b}\right)^2 + p_0^2}, \tag{7}$$

where p_0, p_1, and p_2 are the parameters of the fit, and b is a constant offset and corresponds to the detection threshold. These parameters can be interpreted as the intrinsic resolution evaluated to 11(1)%, the stochastic term, and the noise term, respectively.

Applying this calibration to the main electron cluster, we obtain the final calibrated energy of the main electron. For all the adjacent clusters, we simply apply the charge correction factor obtained from the main electron cluster, assuming that these clusters are gamma-induced "blips" and therefore have a similar response to the main electron. fig. 26 shows the smearing matrices of the main electron and gamma energies after the

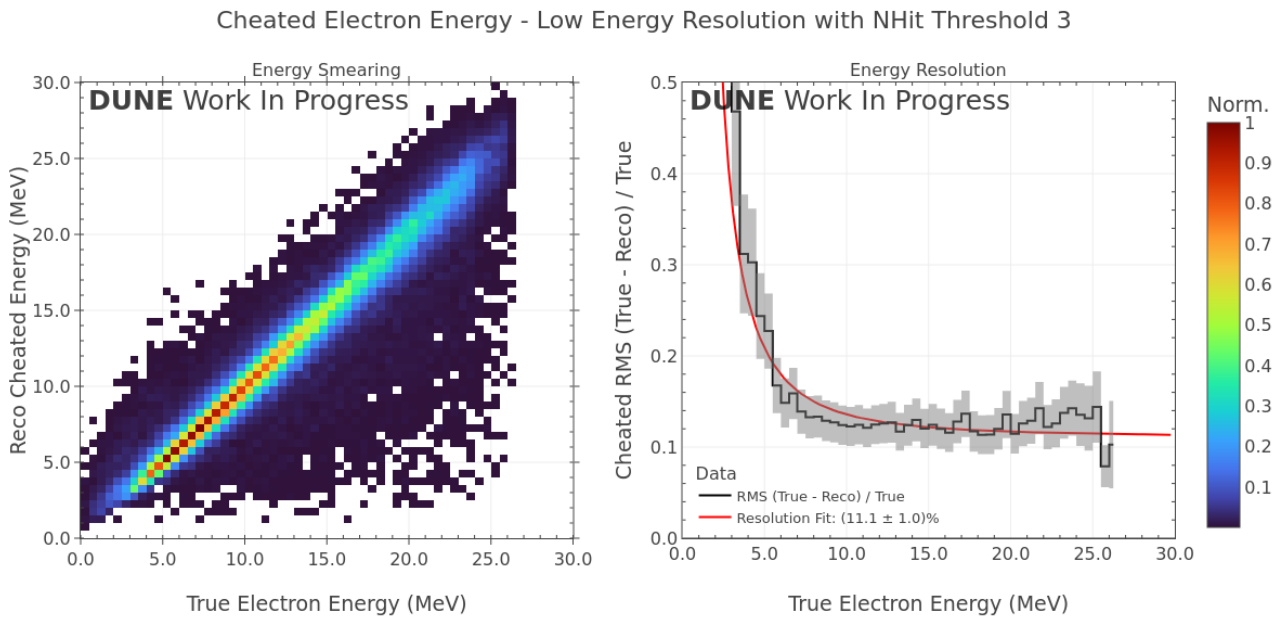


Figure 25: Energy resolution from calibrated charge with cheated clustering and hit threshold of 3.

calibration. Additionally, the smearing matrix of the visible energy (defined as the sum of the main electron and gamma energy) is shown.

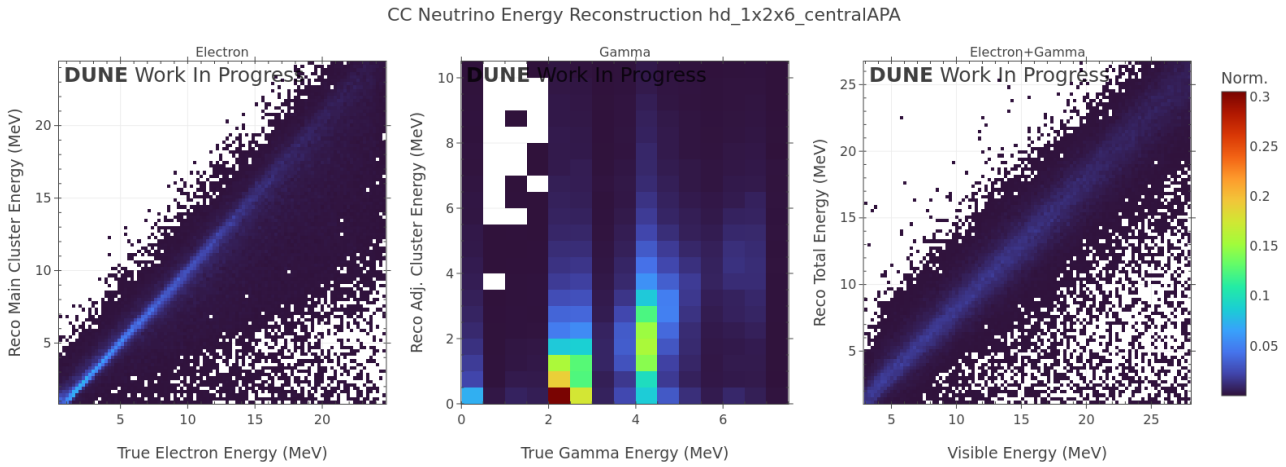


Figure 26: Signal daughter smearing obtained after main electron calibration.

Electron energy resolution with and without flash-matching.

4.4 Neutrino Energy Resolution

The energy resolution is defined as the ratio of the standard deviation of the reconstructed neutrino energy to the mean value of the neutrino energy. The energy resolution is calculated from the calibrated neutrino energies. The results are shown in fig. 27 for the 4 different reconstruction algorithms.

As expected, the **ClusterEnergy** algorithm has the worst energy resolution for neutrino energies below 10 MeV, as it does not take into account the energy deposited by adjacent gamma blips which can amount to more than 50% of the initial neutrino energy. The **SelectedEnergy** and **SolarEnergy** algorithms have the best resolution in this range. As the energy of the neutrino goes higher, the relative weight of the adjacent background clusters starts becoming less important. Over the range of 10 MeV and 20 MeV all algorithms perform almost identical and for the 20 MeV and 30 MeV neutrino energies, the **TotalEnergy** algorithm starts to outperform the others. Importantly, this trend is maintained for any hit threshold selected, ensuring that the energy resolution is robust against the clustering criteria.

Overall, the **SolarEnergy** algorithm is the best choice for neutrino energies below 20 MeV, as it has better (or compatible) energy resolution compared to the other algorithms. The benefit of this method that is not

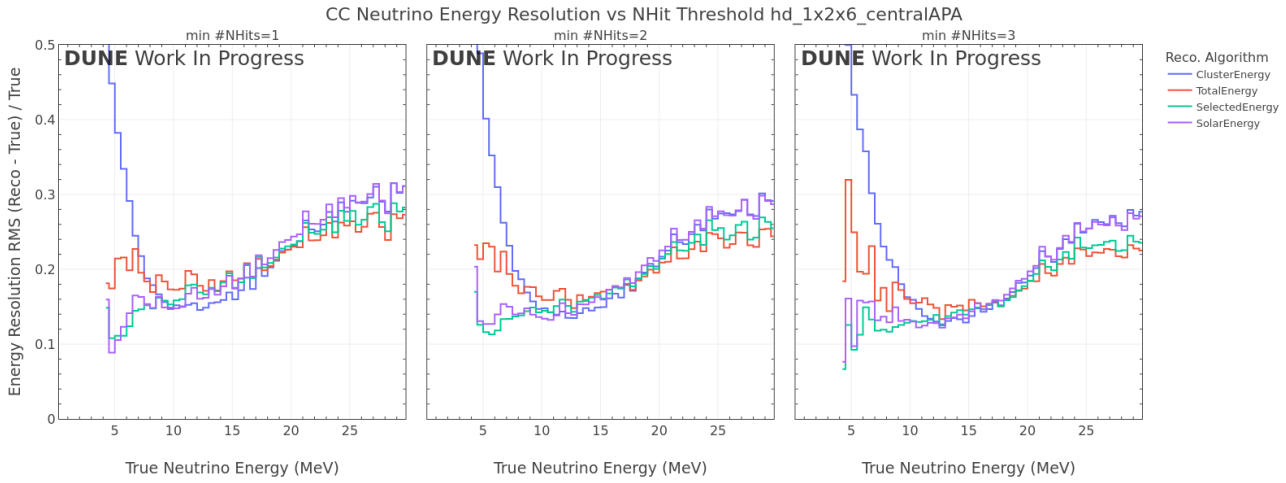


Figure 27: Energy resolution from calibrated neutrino energies with different reconstruction algorithms.

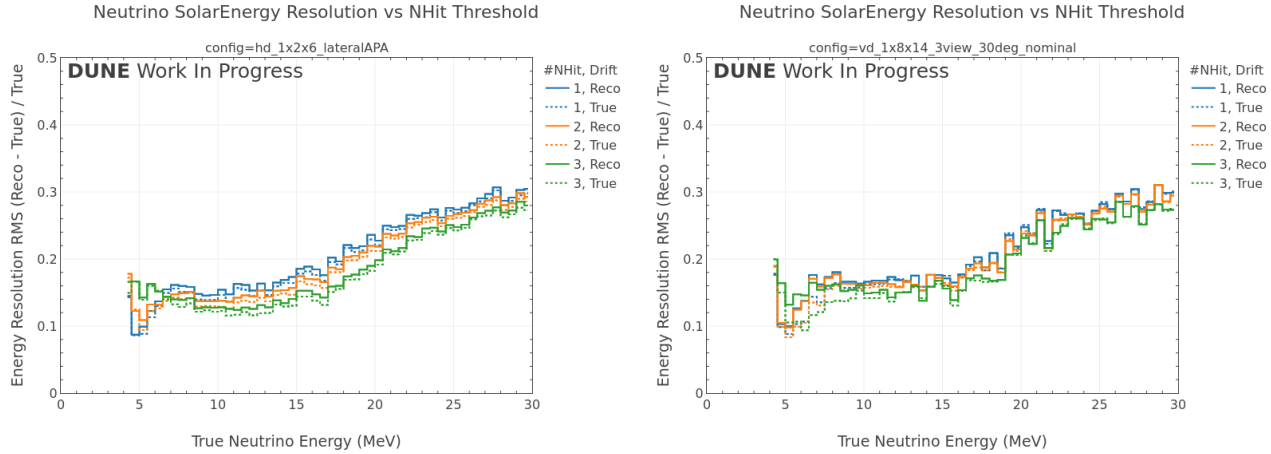


Figure 28: (left) HD (right) VD. Neutrino energy resolution with and without flash–matching. The dashed lines represent the ideal case where the drift coordinate is taken from truth information.

shown in this comparison is that it sets an upper limit to the reconstructed energies of background particles. If not constrained, cavern backgrounds such as neutrons or gammas can contribute significantly with fake events to physics analyses such as the solar neutrino oscillation studies. By only adding a maximum fixed amount of energy to the main clusters produced by these particles, one ensures that their contribution is limited to lower energies compared to the full scope of the solar neutrino spectrum.

Add information about attenuation in the medium and how it affects the energy resolution. Add electron energy resolution.

Add information about the vertex resolution for abs(X) ¿ 5cm and ¡ 15 cm.

4.5 Flash–Matching Effects on Resolution

As already explained, the flash–matching step in the reconstruction process allows to associate a TPC clusters with the corresponding PDS flash and hence obtain the correct drift coordinate of the interaction. In the context of energy reconstruction, this has an impact on the drift attenuation correction on a cluster-by-cluster basis.

As can be seen, in fig. 28, the flash–matching step has a residual effect on the energy resolution. The difference between the flash–matched and non-flash–matched cases are small, indicating that the inefficiencies in the energy reconstruction are dominated by other factors, such as clustering inefficiencies and intrinsic energy losses due to nuclear effects and undetected particles. The dashed lines represent the ideal case where the drift coordinate is taken from truth information, showing that even in this idealized scenario, the resolution does not significantly improve.

5 Conclusion

In this technical note, we have presented the reconstruction algorithms for the DUNE FD, focusing on the neutrino energy and vertex resolution. The reconstruction is based on the TPC cluster vertex reconstruction and flash–matching, which allows us to extract the calibrated neutrino energy and vertex position.

5.1 DUNE Geometry Comparison

So far we have used as reference the *hd_1x2x6_centralAPA* geometry, corresponds to the inner drift volumes of the HD detector. In this section, we compare the results with the *vd_1x8x14_3view_30deg* geometry as well as different background scenarios (nominal, optimistic and shielded), which are described in section 2.1.

In the case of the vertex resolution, we can use the previously defined figure of merit and quote the percent of events with a vertex error below a certain threshold. The results are shown in table 3.

	Resolution (cm) – Sample Size in (1, 3 and 5)σ (%)		
Geometry	**X**	**Y**	**Z**
hd_1x2x6	–	1.1(1) – 59, 91 and 95	1.1(1) – 59, 92 and 96
hd_1x2x6_centralAPA	–	1.1(1) – 58, 90 and 93	1.1(1) – 59, 91 and 95
hd_1x2x6_lateralAPA	–	1.1(1) –	1.1(1) –
vd_1x8x14_3view_30deg	1.3(1) – 32, 49 and 51	1.1(1) – 56, 87 and 91	1.1(1) – 56, 87 and 91
vd_1x8x14_3view_30deg_nominal	2.3(1) – 40, 69 and 90	1.1(1) – 57, 88 and 91	1.1(1) – 59, 91 and 94

Table 3: Vertex resolution comparison between geometries. Percentage of 10 MeV sample below coordinate tolerance.

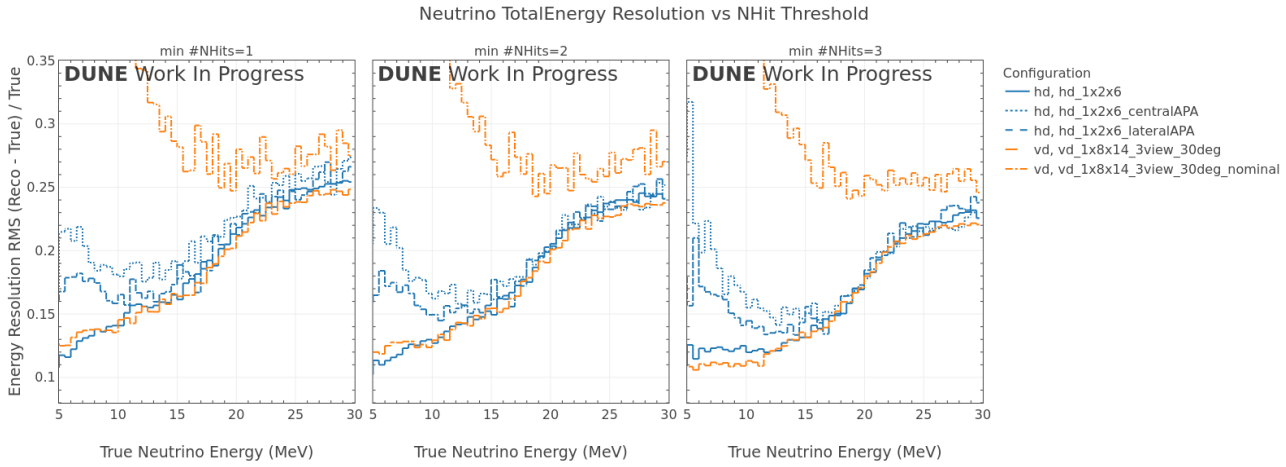


Figure 29: Energy resolution comparison.

Applying the **TotalEnergy** reconstruction algorithm, we can see that the energy resolution is better in the hd_1x2x6_centralAPA geometry, which is expected since it has a lower background rate and smaller drift volume causing therefore less diffusion. The energy resolution is shown in fig. 29.

Applying the **SolarEnergy** reconstruction algorithm we can mitigate this difference in flavour of the vd_1x8x14_3view_30de geometry, which has a lower TPC threshold and is able to pitch up more resolution on the primary cluster. The energy resolution is shown in fig. 30.

5.2 Summary

The vertex and energy reconstruction of low–energy neutrino interactions in DUNE has been studied. The vertex resolution was evaluated using a point–like reconstruction approach that combines the TPC cluster vertex with the associated PDS flash. The achieved vertex resolution is between 0.5 cm and 3.5 cm, depending on the coordinate and the presence of background events. There are significant differences in fiducialization performance between the HD and VD geometries, with the HD geometry achieving higher fiducialization efficiency due to its lower background rates. The energy reconstruction was performed using four different algorithms to sum the energy of reconstructed TPC clusters. The performance of these algorithms was evaluated in terms of energy resolution, fiducialization efficiency, and the impact of flash–matching on the energy reconstruction. Contrary to the vertex resolution, the energy resolution does not show significant differences between the two geometries

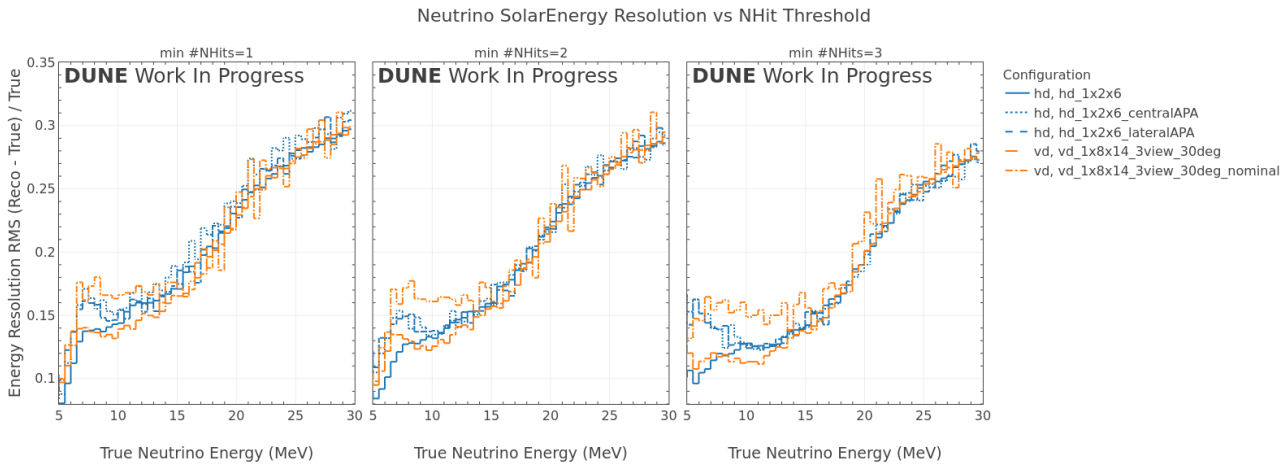


Figure 30: Energy resolution comparison.

provided that one choses the appropiate reconstruction algorithm. The best energy resolution is achieved using the *SolarEnergy* algorithm, which yields a resolution of approximately 20% at 15 MeV and improves to about 10% at 5 MeV. The flash–matching step has a minor impact on the energy resolution, indicating that other factors, such as clustering inefficiencies and intrinsic energy losses, dominate the resolution. Overall, the reconstruction algorithms demonstrate good performance for low–energy neutrino interactions in DUNE, with potential for further optimization and improvement.

Based on the amount of background activity in the VD module, other reconstruction objectives such as channel tagging (out of the scope of this work) and particle identification are likely to be more challenging than in the HD module. The case of a shielded VD module, which has background rates comparable to the HD module, is therefore highly encouraged and expected to yield similar low–energy reconstruction performance.

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List of Acronyms

APA Anode Plane Assembly

CC Charged Current

CPA Cathode Plane Assembly

CRP Charge Readout Plane

DUNE Deep Underground Neutrino Experiment

ES Elastic Scattering

FD Far Detector

HD Horizontal–Drift

LAr Liquid Argon

LArSoft Liquid Argon Software Framework

LArTPC Liquid Argon Time Projection Chamber

MARLEY Model of Argon Reaction Low Energy Yields

MC Monte Carlo

ML Machine Learning

NC Neutral Current

PCB Printed Circuit Board

PDS Photon Detection System

PE Photoelectron

RMS Root Mean Square

SURF Sanford Underground Research Facility

TPC Time Projection Chamber

VD Vertical–Drift

XA X–ARAPUCA

6 Appendix

Introduction

Due to its nature as a noble gas, ^{222}Rn is difficult to remove from the detector environment, making it a persistent background source also in detector components. Among its progeny, several isotopes are particularly important:

- **Bismuth (^{214}Bi)**: A short–lived beta emitter with a high Q–value (up to 3.27 MeV) and multiple gamma lines. It is the most significant source of high–energy background from ^{222}Rn [7].
- **Bismuth (^{210}Bi)**: A beta emitter that produces a continuum low–energy background up to 1.16 MeV.
- **Lead (^{210}Pb)**: A long–lived isotope ($t_{1/2} = 22.3$ years) that accumulates on detector surfaces [8].
- **Polonium (^{210}Po)**: An alpha emitter with a half–life of 138 days. It can be used as a surface contamination proxy due to its distinct alpha signal [9].